NUCLEAR HORIZONS

(A Bulletin of the Indian Nuclear Physics Association)

ADVANCING FUNDAMENTAL AND APPLIED NUCLEAR SCIENCE
FOR A SELF-RELIANT INDIA

Volume 1 | Issue 1 | Year 2026

Aligned with Atmanirbhar Bharat

INAUGURAL ISSUE

PUBLISHED BY

INDIAN NUCLEAR PHYSICS ASSOCIATION (INPA)

JANUARY 2026

Registration No.: VAR/00818/2026-27

ISSN: Applied for

Published by:
Indian Nuclear Physics Association
(Scientific Society)

Disclaimer

Contact Information

Indian Nuclear Physics Association (INPA)
Email: inpa.bharat@gmail.com
Website: Under preparation

Advisory Committee

(Indian Nuclear Physics Association)

The Indian Nuclear Physics Association (INPA) is pleased to constitute its **Advisory Committee**, comprising eminent scientists and distinguished academicians who have made outstanding contributions to the growth and advancement of nuclear physics in India. The Committee provides strategic guidance and intellectual leadership to help shape the vision, activities, and long-term roadmap of the Association. The Advisory Committee will play a crucial role in advising INPA on emerging research directions, national and international collaborations, capacity building, and outreach initiatives, in alignment with **Vision-2035 for Nuclear Science and Technology**.

Members

1. **Prof. P. C. Sood**
   Padma Shri Awardee (2023)
   Banaras Hindu University, Varanasi
2. **Prof. S. C. Pancholi**
   University of Delhi, Delhi
3. **Prof. R. Prasad**
   Aligarh Muslim University, Aligarh
4. **Prof. R. K. Bhandari**
   Variable Energy Cyclotron Centre, Kolkata
5. **Prof. V. N. Bhoraskar**
   Savitribai Phule Pune University, Pune
6. **Prof. S. Kailas**
   Bhabha Atomic Research Centre, Mumbai
7. **Prof. Amit Roy**
   Inter-University Accelerator Centre, New Delhi
8. **Prof. S. K. Singh**
   Aligarh Muslim University, Aligarh
9. **Prof. R. K. Choudhury**
   Bhabha Atomic Research Centre, Mumbai
10. **Prof. D. K. Srivastava**
    Variable Energy Cyclotron Centre, Kolkata
11. **Dr. Dinakar Kanjilal**
    Inter-University Accelerator Centre, New Delhi
12. **Prof. Avinash C. Pandey**
    Inter-University Accelerator Centre, New Delhi
13. **Prof. A. K. Sinha**
    UGC–DAE Consortium for Scientific Research
14. **Dr. D. K. Mohapatra**
    Atomic Energy Regulatory Board, Mumbai
15. **Prof. N. M. Badiger**
    Karnatak University, Dharwad, Karnataka
16. **Prof. V. K. B. Kota**
    Physical Research Laboratory, Ahmedabad
17. **Prof. Bijay K. Agrawal**
    Saha Institute of Nuclear Physics, Kolkata
18. **Prof. Radhey Shyam**
    Saha Institute of Nuclear Physics, Kolkata
19. **Dr. R. P. Singh**
    Inter University accelerator Center, New Delhi
20. **Dr. S. Muralithar**
    Inter University accelerator Center, New Delhi

The Indian Nuclear Physics Association (INPA) gratefully acknowledges the guidance and support of the Advisory Committee. Their expertise will strengthen research, education, collaboration, and outreach in nuclear science, and help position INPA as a national platform of excellence.

Executive Committee
(Indian Nuclear Physics Association)

President
Prof. A. K. Jain – Indian Institute of Technology, Roorkee

Vice Presidents
Prof. Samit Mandal – University of Delhi, Delhi
Prof. R. Palit – Tata Institute of Fundamental Research, Mumbai, Maharashtra

Executive Secretary
Prof. B. P. Singh – Aligarh Muslim University, Aligarh, U.P.

Joint Secretaries
Prof. Manoj Kumar Sharma (Joint Secretary–I) – TIET, Patiala, Punjab
Prof. Bivas R. Behera (Joint Secretary–II) – Panjab University, Chandigarh, Punjab

Treasurer
Prof. Ajay KumarTyagi – Banaras Hindu University, Varanasi, U.P.

Members
Prof. N. L. Singh – M. S. University of Baroda, Vadodara
Prof. Manoj Kumar Sharma – University of Lucknow, Lucknow
Prof. Tarkeshwar Trivedi – University of Allahabad, Prayagraj
Dr. Mohd. Shuaib – Aligarh Muslim University, Aligarh

Council Members
Prof. S. Santra – BARC, Mumbai
Prof. K. P. Singh – Panjab University, Chandigarh
Prof. M. M. Musthafa – University of Calicut, Calicut
Dr. Sandeep Ghugre – UGC-DAE Consortium for Scientific Research, Kolkata
Prof. Tilak Ghosh – VECC, Kolkata
Dr. Subir Nath – IUAC, New Delhi
Prof. H. M. Mittal – NIT, Jalandhar
Prof. Harleen Dahiya – NIT, Jalandhar
Prof. M. Balasubramaniam – Bharathiar University, Coimbatore
Prof. J. J. Das – Cotton University, Assam
Prof. Prince Ganai – NIT, Srinagar, Kashmir
Prof. Sunita Gupta – Agra College, Agra
Prof. Avinash Agrawal – Bareilly College, Bareilly
Dr. Rhine Kumar – Cochin University of Science and Technology, Kochi
Dr. Dipika Patel – SVNIT, Surat, Gujarat
Dr. Akashrup Banerjee – SINP, Kolkata
Prof. Indranil Mazumdar – TIFR, Mumbai
Prof. A. M. Vinod Kumar- University of Calicut, Calicut
Prof. Javid Sheikh, Kashmir University, Srinagar, Jammu and Kashmir
Dr. Sunethra Ramanan, Department of Physics, IIT Madras.

काशी हिन्दू विश्वविद्यालय
BANARAS HINDU UNIVERSITY

अजित कुमार चतुर्वेदी
कुलपति
Ajit Kumar Chaturvedi
Vice-Chancellor

April 16, 2026

MESSAGE

It gives me immense pleasure to note that Banaras Hindu University is serving as the host institution for the launch and office of the Indian Nuclear Physics Association (INPA). This initiative reflects a strong commitment to nurturing the next generation of scientific leaders from the nuclear physics fraternity. The formation of INPA is in alignment with *Mega Science Vision–2035: Nuclear Physics*. It is indeed a matter of pride for the University to be associated with such a significant national endeavour.

The inaugural volume of the Association's bulletin, *Nuclear Horizons: A Bulletin of the Indian Nuclear Physics Association*, highlights nuclear facilities and activities across universities and research institutions. The bulletin aims to create awareness among students, teachers, and the general public about the importance and applications of nuclear science.

As the host institution, Banaras Hindu University remains committed to providing all possible academic and administrative support to INPA in achieving its goals. I am sure the INPA will help in advancing excellence in nuclear science and technology in the country.

Furthermore, INPA should actively mobilize resources to support student research, laboratory development, and to provide early-career scientists with mentorship, opportunities, and the resources necessary to thrive in their research careers.

On behalf of the Banaras Hindu University, I extend my warm greetings and best wishes for the grand success of the Indian Nuclear Physics Association (INPA). I also congratulate everyone associated with the establishment of this Association and the publication of the inaugural issue of its bulletin.

I wish INPA every success in its mission to advance Nuclear Physics in India and contribute meaningfully to the nation's scientific progress.

Chaturvedi

(**Ajit Kumar Chaturvedi**)

CONTENTS

President's Message

Dear Young Friends and Esteemed Colleagues,

I take this opportunity to wish you all a very happy New Year 2026. May this year be the beginning of the fulfilment of our aspirations and prove fruitful in opening new pathways for the growth of the nuclear physics community.

It gives me great pleasure to write this note addressing the nuclear physics community of India. It has been a long-cherished dream of many of us to have a cohesive platform that will act as the collective voice of the small but vibrant nuclear physics community. Such a platform will fulfil our aspirations, guide the community into future developments in this strategic area, and also highlight the concerns of the community. So far, the needs and problems of the nuclear physics community, although talked about, have remained largely ignored. This is particularly true for the hundreds of nuclear scientists working in universities, colleges, and other such institutions. Nuclear physics, by nature, is one of the toughest areas of physics and requires years of work to publish even a single good paper. The development of a simple laboratory setup often takes years to fructify. Yet all this hard work done within the country receives little or no recognition. As a result, we continue to lose knowledge and expertise in one of the most critical and challenging areas of physics.

Overcoming several false starts of the past, a national online meeting of the nuclear physics community in India was finally held on 5th November 2025, with the active support of Prof. B. P. Singh (AMU). More than 100 participants formally joined and initiated the formation of the Indian Nuclear Physics Association (INPA). This initiative is in line with one of the key recommendations of the Mega Science Vision–2035: Nuclear Physics document released by the Principal Scientific Advisor (PSA) to the Government of India and made public on 14th September 2023. As advised by the PSA, Prof. A. K. Sood, the Indian nuclear physics community should organize itself and take the necessary steps to turn the vision outlined in this report into reality. The national policy document calls for the strengthening of nuclear science, recommending upgraded accelerator facilities, detector systems, theory centres, HPC clusters, and underground laboratories in nuclear physics. However, no formal structure exists to undertake this enormous exercise.

We are fortunate that this initiative has been blessed by senior nuclear physicists across the country from universities, IITs, and DAE institutions. More than 200 members have already expressed their willingness to join INPA. It is now up to us to turn the community's vision into reality. More than two years have already passed since the release of the report, and no action has yet been taken. I feel that if we do not act now, nobody will. Today, nuclear physics courses are being phased out or deleted from institutions of higher learning. Top-level institutes created for and by nuclear physicists refuse to recruit and sustain even small nuclear physics groups. Some newly created science institutes have openly declared that nuclear physics will not enter their gates. We have become accustomed to shining in the reflected glory of work done in offshore laboratories and to following fashionable research trends that make headlines abroad. While science is indeed a global endeavour, it should not come at the cost of depleting our in-house knowledge and strength. We need to strive for a balance between the two.

"The Vision–2035 document, prepared after long consultations and intense discussions, presents us with an opportunity to move forward in the right direction. INPA should help fulfil future aspirations and address the rapidly changing landscape in this strategically important field. INPA should also strive to place the concerns of the younger generation, who are choosing nuclear physics as a profession with great passion, at the forefront".

As the founding President of INPA, I will strive to fulfil these obligations to the best of my ability. In doing so, I will need the full support of every member. We will hold frequent consultations and seminars to keep our younger colleagues fully engaged. Prof. B. P. Singh has curated the first issue of the Bulletin of INPA beautifully, and I congratulate him on this significant effort and its timely publication. I also welcome the first Executive Committee announced in this issue, which is only a starting point. Once the society is formally registered, we will be able to implement the constitution of the association. We remain fully committed to the aims of the society and will keep our eyes and ears open to suggestions in this regard. Wishing you all the very best, and with regards.

Ashok Jain

A. K. Jain

From the Chief Editor's Desk

Indian Nuclear Physics – Vision 2050

It is a matter of great pride to present the First Issue of the "Nuclear Horizons"- A bulletin of Indian Nuclear Physics Association (INPA), the official voice of the emerging nuclear physics community. This inaugural volume reflects a collective aspiration of the nuclear physics community in India, an aspiration that took a decisive step forward during the national online meeting held on 5 November 2025, where over a hundred members came together with a shared purpose: *to organise, strengthen, and rejuvenate nuclear physics in the country*. Towards the end of the meeting, Prof. A. K. Jain was nominated as the President, and Prof. B. P. Singh was nominated as the Executive Secretary of INPA.

The formation of INPA is firmly grounded in the vision laid out in the **Mega Science Vision – 2035 (Nuclear Physics)** document released by the Office of the Principal Scientific Advisor in 2023. This national roadmap clearly emphasises the need to build modern accelerator facilities, advanced detector systems, theory centres, high-performance computing clusters, and underground laboratories. Crucially, it also recommends the establishment of a national society to unify and coordinate the Indian nuclear physics community. INPA is the community's response to this call.

We sincerely thank all the Advisory Committee members for kindly accepting our invitation and for their valuable support in all our endeavours. We also extend our gratitude to all the other Executive Committee members for their continued cooperation and contributions. We look forward to their continued cooperation and support in building INPA as an important and effective association for the development of the subject and the scientific community, thereby contributing to national development.

This INPA bulletin, "***Nuclear Horizons***", is conceived as a platform to articulate this vision and document our journey. The discussions during the launch meeting highlighted several urgent concerns, declining faculty positions, shrinking research infrastructure, reduced laboratory training, and inadequate representation in academic bodies. These challenges threaten the vitality of nuclear physics in India. A field that is central not only to scientific discovery but also to national strategic capability, medical innovation, energy solutions, and advanced technologies. The unanimous decision to form INPA reflects a clear conviction that "India needs a strong, organised, and forward-looking nuclear physics community". The association aims to advocate for upgraded facilities and fair opportunities, support education and training, build national databases, facilitate collaborations, and engage actively with policy-making agencies. A major emphasis will be on empowering young researchers through mentoring, internships, and training schools.

As Chief Editor, it is a matter of great pride to present this inaugural issue, which marks the beginning of a collective journey for the Indian Nuclear Physics Association (INPA). *Nuclear Horizons*—the official bulletin of INPA—seeks to capture the aspirations, ideas, and momentum of a unified nuclear physics community, while documenting scientific progress and shaping dialogue on the future of nuclear physics in India.

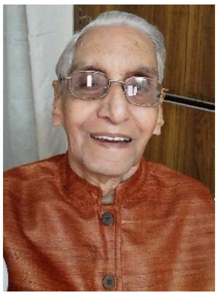
I gratefully acknowledge Prof. A. K. Jain for his leadership and foresight in initiating the formation of INPA, inspired by the vision articulated in the **Mega Science Vision – 2035** document. I also extend my sincere appreciation to all colleagues and contributors whose support, commitment, and shared belief in the strength of collective action have transformed this vision into a meaningful national initiative.

This is only the beginning. With sustained community engagement, transparent governance, and a shared commitment to excellence, INPA will emerge as a strong national platform leading India toward its Nuclear Physics Vision 2050.

I invite all members of the community, senior scientists, young researchers, educators, students, and stakeholders, to participate actively, contribute generously, and help shape a brighter future for Indian nuclear physics. Last but not the least, feel free to reach out to any of us, and contribute to the endeavour and also contribute to the next issues of the INPA bulletin.

B. P. Singh

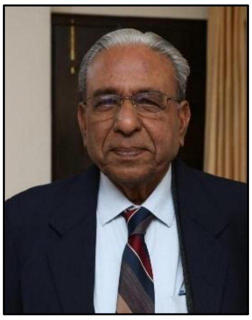
Messages from Senior Nuclear Physicists



Prof. P. C. Sood, recipient of the Padma Shri award, 2023

Prof. Prakash Chandra Sood (born 1928) is a distinguished nuclear physicist from Punjab, India. He is widely respected for his significant contributions to nuclear physics and for his long association with Banaras Hindu University (BHU). Prof. Sood served as the Head of the Department of Physics in 1969 and was instrumental in establishing the first Van de Graaff Accelerator at BHU, marking a major milestone in experimental nuclear physics in India. In recognition of his lifelong dedication to science and education, he was conferred the Padma Shri in 2023. In his message to the community, Prof. Sood expressed his warm support and encouragement, through following message:

> ***"I am delighted to learn about this move of the Nuclear Physics Community to have a united forum to move forward. I shall be honoured to be a part of this move in any way the Community desires." Further, "In an evolving venture like ours, role of superannuated seniors is to provide direction and appropriate solid foundation for the FORWARD-LOOKING GEN-NEXT SCHOLARS to take it up from there to the desired level of excellence. Let us all work as a well-knit team in this path breaking task".***

Prof. S. C. Pancholi and Prof. R. K. Bhandari

We believe that the future of Nuclear Physics in our country will be strengthened through the collective efforts of this organized body. The formation of this platform fulfils a long-felt need of the community by bringing together researchers, educators, and professionals to enhance collaboration, advance research, and support national scientific development. Further details are presented in our article, "Promotion of Nuclear Physics with Advanced Funding for Teaching and Research in the University System in the Country — A Much-Needed Action Plan for Nuclear Physics."

Homi Bhabha's Legacy in India's Nuclear Vision

D. K. Mohapatra
Outstanding Scientist & Director, TS&RD
Directorate of Technical Support & Regulatory Documents
Atomic Energy Regulatory Board, Mumbai 400094
Email: dir.dtsrd@aerb.gov.in

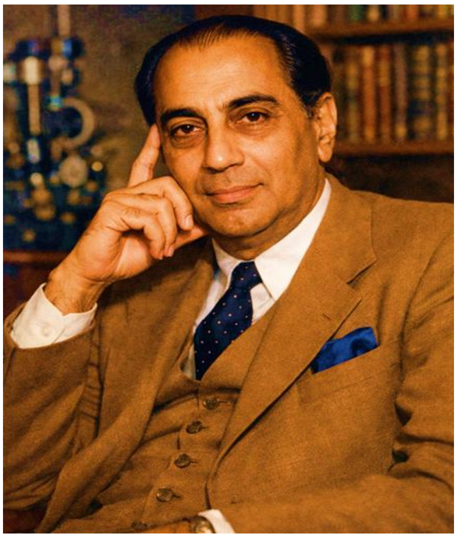
Abstract: *Dr. Homi Jehangir Bhabha was the principal architect of India's nuclear vision, seamlessly integrating frontier scientific research with long-term national development goals. At a time of limited resources, he laid the foundations of a self-reliant atomic energy programme by building world-class institutions such as TIFR, BARC, and the Department of Atomic Energy. His foresight led to the formulation of India's three-stage nuclear power programme, tailored to the country's uranium and thorium resources. Beyond power generation, Bhabha emphasized peaceful applications of nuclear science in medicine, agriculture, and industry. His legacy continues to guide India's pursuit of energy security, technological sovereignty, and sustainable development.*

Introduction

The development of nuclear energy in India is inseparable from the vision and leadership of Dr. Homi Jehangir Bhabha, who was not only the principal architect of India's atomic energy programme but also one of the foremost institution builders in the history of modern Indian science. His legacy lies in the integration of frontier scientific research with long-term national policy objectives, particularly in the domain of nuclear energy. At a time when the country lacked both industrial capacity and advanced scientific infrastructure, Dr. Bhabha conceptualised a nuclear programme that was technologically ambitious, institutionally integrated, and strategically aligned with India's long-term developmental needs. His contribution lay not merely in initiating nuclear research, but in creating an enduring scientific, organisational, and policy-driven framework within which nuclear energy could evolve as a national enterprise.

Academic Formation and Scientific Contributions

Homi Bhabha, born on 30 October 1909 into an intellectually distinguished Parsi family, received his early education in Bombay (presently known as Mumbai), which exposed him to both scientific and cultural pursuits, thereby laying the foundation for a lifelong synthesis of science, art, and humanism. In 1927, he joined Gonville and Caius College, University of Cambridge, and completed both the Engineering Tripos and the Mathematics Tripos with first-class distinctions.



Homi Jehangir Bhabha
(30 October 1909 – 24 January 1966)

His doctoral research under Prof. Ralph H. Fowler at Cambridge focused on quantum electrodynamics and cosmic ray phenomena, fields that were at the cutting edge of physics at the time. His work on the cascade theory of electron showers, carried out in collaboration with Walter Heitler, provided a quantitative explanation of the soft component of cosmic radiation and remains a cornerstone of astro-particle physics. Dr. Bhabha also made pioneering theoretical predictions regarding the properties and decay of mesons, including early insights into relativistic time-dilation effects in muon decay. Another major contribution, now universally known as Bhabha scattering, addressed the interaction between electrons and positrons and continues to be of practical importance in the calibration of particle-accelerator experiments.

The Institution Builder: *from TIFR to Atomic energy programme*

The outbreak of the Second World War in 1939 brought Bhabha back to India, where he joined the Indian Institute of Science (IISc), Bengaluru, at the invitation of Sir C. V. Raman. What began as a temporary stay evolved into a decisive commitment to building scientific capability in his home country. Bhabha became increasingly convinced that India's future depended on the creation of institutions capable of sustaining world-class research.

In 1944, he articulated this vision in a landmark proposal to the Sir Dorabji Tata Trust, arguing that advanced research in fundamental physics was essential not only for intellectual reasons but also for long-term technological and industrial development. This proposal led to the establishment of the Tata Institute of Fundamental Research (TIFR) in 1945.

TIFR soon became the nucleus from which India's atomic energy programme evolved. In 1948, the Atomic Energy Commission was constituted, with Bhabha as its founding Chairman, and the Department of Atomic Energy (DAE) was established with him as its Secretary. Bhabha's vision for atomic energy was comprehensive. He viewed nuclear science as a strategic resource for power generation, medicine, agriculture, and national security, while strongly advocating peaceful applications. The Atomic Energy Establishment at Trombay, later renamed the Bhabha Atomic Research Centre (BARC), was founded in 1954 as a multidisciplinary laboratory encompassing nuclear physics, reactor engineering, radiochemistry, health physics, metallurgy, electronics, etc. Dr. Bhabha established the BARC Training School in 1957 to cater to the manpower needs of the expanding atomic energy research and development programme.

In Bhabha's own words, "When nuclear energy has been successfully applied for power production in, say, a couple of decades from now, India will not have to look abroad for its experts but will find them ready at hand." Dr. Bhabha emphasized self-reliance in all fields of nuclear science and engineering.

Research Reactors in India

A defining feature of Dr. Bhabha's strategy was the early emphasis on research reactors. Rather than relying solely on theoretical studies or imported designs, he insisted on hands-on experience in reactor construction and operation as the basis for technological self-reliance.

India's first reactor, APSARA, a swimming-pool-type research reactor, achieved criticality on 4 August 1956. APSARA provided a practical platform for studying reactor physics, neutron behaviour, and radiation safety. It also supported radioisotope production, thereby linking nuclear research to societal applications in medicine and industry.

This was followed by CIRUS, a 40 MWth heavy-water-moderated research reactor built in collaboration with Canada, which was commissioned on 10 July, 1960. CIRUS was extensively used for condensed-matter research using neutron beams, material irradiation, fuel testing, neutron activation analysis, and the production of radioisotopes for applications in medicine, agriculture, and industry. CIRUS provided the foundation for understanding the development of technologies related to natural uranium-based fuel, heavy water, and reactor systems, which eventually paved the way for the design and development of Pressurised Heavy Water Reactors (PHWRs) that became the backbone of the Indian nuclear power programme.

During subsequent years, the country witnessed the design and commissioning of several advanced research reactors, such as

ZERLINA, DHRUVA, and PURNIMA-I, II, and III. In fact, PURNIMA-III was a U-233-based mock-up reactor facility at BARC that led to the design of the 30 kW Kalpakkam Mini Reactor (KAMINI), which has been operating at the Indira Gandhi Centre for Atomic Research, Kalpakkam, since 1996. On similar lines, following the permanent shutdown of the APSARA reactor in 2009, a new and advanced version was commissioned at BARC, named APSARA-Upgraded (APSARA-U). This reactor uses indigenously developed low-enriched uranium in the form of uranium silicide and provides a high neutron flux of the order of 6.1×10^{13} n/cm^2/s for a wide range of applications, including isotope production, neutron activation analysis, neutron radiography, detector testing, and radiation shielding studies.

Collectively, these research reactors facilitated the training of engineers and operators, validated indigenous design methodologies, and generated critical data for the development of power reactors in the country.

The Three Stage Nuclear Power Programme: *A strategic foresight*

Among Bhabha's most far-reaching contributions was the formulation of a long-term nuclear power strategy tailored to India's natural resource profile. Recognising the country's limited uranium reserves and substantial thorium deposits, he articulated a three-stage nuclear power programme that sought to progressively enhance fuel utilisation and ensure long-term sustainability.

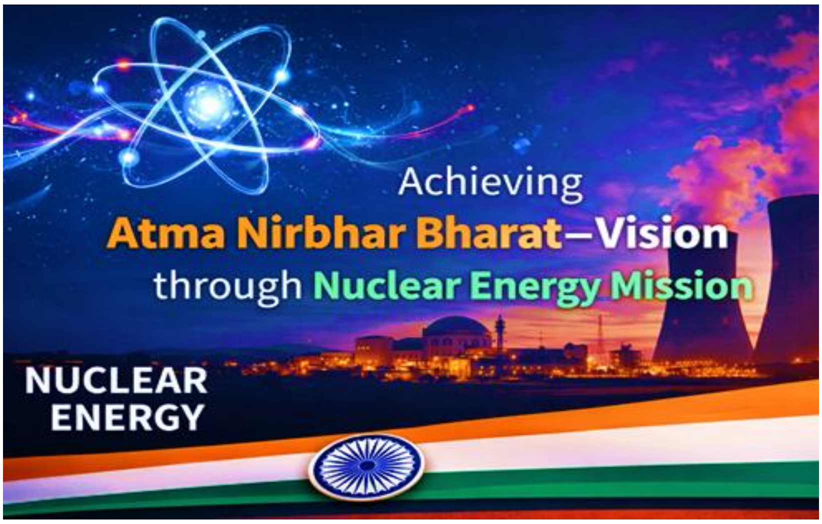
The first stage envisaged the deployment of Pressurised Heavy Water Reactors (PHWRs) for electricity generation, which use natural uranium (UO_2) containing 0.7% fissile U-235 and 99.3% U-238 as fuel, and heavy water (D_2O) as the moderator and primary coolant. The second stage involved Fast Breeder Reactors (FBRs), in which the plutonium extracted from spent PHWR fuel serves as the seed fuel. In an intricate design, the FBR not only consumes plutonium but also breeds additional fissile material, thereby expanding the nuclear fuel base. The second stage also aims to harness India's abundant thorium reserves through FBRs that are capable of converting Th-232 into fissile U-233. It is envisaged that U-233 would drive the futuristic third stage of the nuclear power programme, deploying either thermal or fast reactor systems.

The Present Nuclear Power Scenario

The Indian nuclear programme today comprises 24 power reactors operating at seven sites, with an installed capacity of 8,780 MWe (megawatt electric). These reactors encompass a wide range of technologies, including Pressurised Heavy Water Reactors (PHWRs), Boiling Water Reactors (BWRs), and Pressurised Water Reactors (PWRs), with varying power capacities. The home-grown PHWRs have evolved from 220 MWe capacity to 540 MWe, and have ultimately reached the standardised 700 MWe design.

Currently, the Nuclear Power Corporation of India Limited (NPCIL), a public sector enterprise under the Department of Atomic Energy (DAE), is constructing seven reactors with a total capacity of 6,100 MWe. These include Rajasthan Atomic Power Project (RAPP) Unit-8 (700 MWe PHWR), Kudankulam Nuclear Power Station (KKNPS) Units-3 to 6 (four units of 1,000 MWe Light Water Reactors), and Gorakhpur Haryana Anu Vidyut Pariyojana (GHAVP) Units-1 and 2 (two units of 700 MWe PHWRs), which are at various stages of construction. Further, the Government of India has sanctioned ten fleet-mode 700 MWe PHWRs, which are expected to lead to a total nuclear power capacity of 21,980 MWe by 2031–32.

The second stage of the nuclear power programme was heralded by the 40 MWth (megawatt thermal) Fast Breeder Test Reactor (FBTR) attaining criticality in October 1985. A unique feature of this reactor is its indigenously developed uranium–plutonium carbide fuel, with liquid sodium as the coolant. With the rich operational experience gained from FBTR operation, along with advances in sodium technology and fuel reprocessing capabilities, the country has embarked upon the design and construction of the 500 MWe Prototype Fast Breeder Reactor (PFBR). Towards the utilisation of



thorium in the third stage, the Advanced Heavy Water Reactor (AHWR) has been conceptualised, incorporating several passive safety systems in its design, thereby enabling it to meet the requirements of the Generation-III+ advanced reactor category.

Societal Applications of Nuclear Technology

Bhabha's vision has enabled the Department of Atomic Energy to emerge as a key pillar of India's scientific and technological ecosystem, with primary responsibility for the design, construction, and operation of nuclear power reactors, supported by comprehensive nuclear fuel cycle capabilities. In parallel, the Department has developed advanced technologies in areas such as accelerators, lasers, high-performance computing, advanced materials, and precision instrumentation.

The wide-ranging applications of radiation technology include improvements in agriculture through better crop varieties and post-harvest technologies; advances in healthcare through radiodiagnosis and radiotherapy, particularly for cancer management; as well as solutions for safe drinking water, environmental protection, and industrial growth.

Concluding Remarks

Dr. Homi Bhabha's untimely death in 1966 marked the end of an era, but not the end of his influence. The institutions he founded, the policies he shaped, and the human capital he nurtured continue to define India's nuclear and scientific enterprise. His long-term vision has fructified into the current National Nuclear Energy Mission for Viksit Bharat, which aims to achieve 100 GW of nuclear power capacity in the country by 2047, to ensure energy security and meet net-zero carbon emission goals. The strategies for this ambitious mission include the development of Small Modular Reactors (SMRs) based on home-grown PHWR and LWR technologies, Molten Salt Breeder Reactors (MSBRs), and private sector participation in nuclear power generation. The MSBR technology would provide a unique pathway to utilise thorium directly to achieve the third stage of the nuclear power programme. In parallel, BARC is also developing the Innovative High-Temperature Reactor (IHTR), with the aim of providing high-temperature process heat for hydrogen production through thermochemical water splitting.

In conclusion, India's nuclear vision remains firmly anchored in the principles articulated by Dr. Homi Bhabha. It reflects faith in fundamental research and confidence in indigenous capability. It also upholds an unwavering commitment to the peaceful and constructive use of atomic energy. This commitment is formally embodied in the Shanti Bill on nuclear energy (Safety, Harmony, and Non-proliferation through Technology Initiative), a legislative assurance that nuclear science and technology in India shall be used exclusively for peaceful purposes and societal development. The Bill reinforces responsible governance, transparency, and public trust in nuclear energy. Together, Bhabha's vision and the

Shanti Bill position nuclear energy as a cornerstone of energy security, sustainability, and technological self-reliance for a Viksit Bharat.

References:

1. Venkataraman, G., Bhabha and His Magnificent Obsessions, Universities Press (1994).
2. Department of Atomic Energy, Government of India website (*www.dae.gov.in*)
3. DAE, Government of India, Annual Report (2023-24)

Prof. D. K. Mohapatra *is a senior nuclear scientist at the Atomic Energy Regulatory Board (AERB), India, currently serving as Director, Technical Support & Regulatory Documents (TS&RD). He has made significant contributions to nuclear safety regulation, technical evaluations, and the development of regulatory guidelines in India.*

India's Nuclear Physics Roadmap to 2035

Based on the Mega Science Vision–2035 (Nuclear Physics) Document

B. P. Singh
Department of Physics, Aligarh Muslim University, Aligarh, U.P.

Introduction

Nuclear physics has long been one of the foundational pillars of India's scientific, technological, and strategic landscape. From fundamental research on nuclear structure and reactions to wide-ranging applications in medicine, agriculture, materials science, space research, and national security, its impact is deeply interwoven with India's developmental goals. Recognising the rapid global advances in accelerator technology, rare-isotope facilities, and computational nuclear science, as well as the transformative potential of nuclear research, the Office of the Principal Scientific Advisor (PSA), Government of India, released the *Mega Science Vision–2035: Nuclear Physics* document in September 2023. This document articulates a bold national strategy aimed at modernising India's nuclear research infrastructure, fostering high-end theoretical and experimental capabilities, strengthening the manpower pipeline, and positioning India among the global leaders of nuclear science by 2035. The formation of the *Indian Nuclear Physics Association (INPA)* in 2025 is a landmark step toward realising this vision, providing a unified platform that aligns the community's efforts with national priorities. This feature article presents an integrated summary of the roadmap. It highlights the scientific motivations, recommended facilities, manpower strategies, and the vision for a strong and vibrant nuclear physics ecosystem over the next decade.

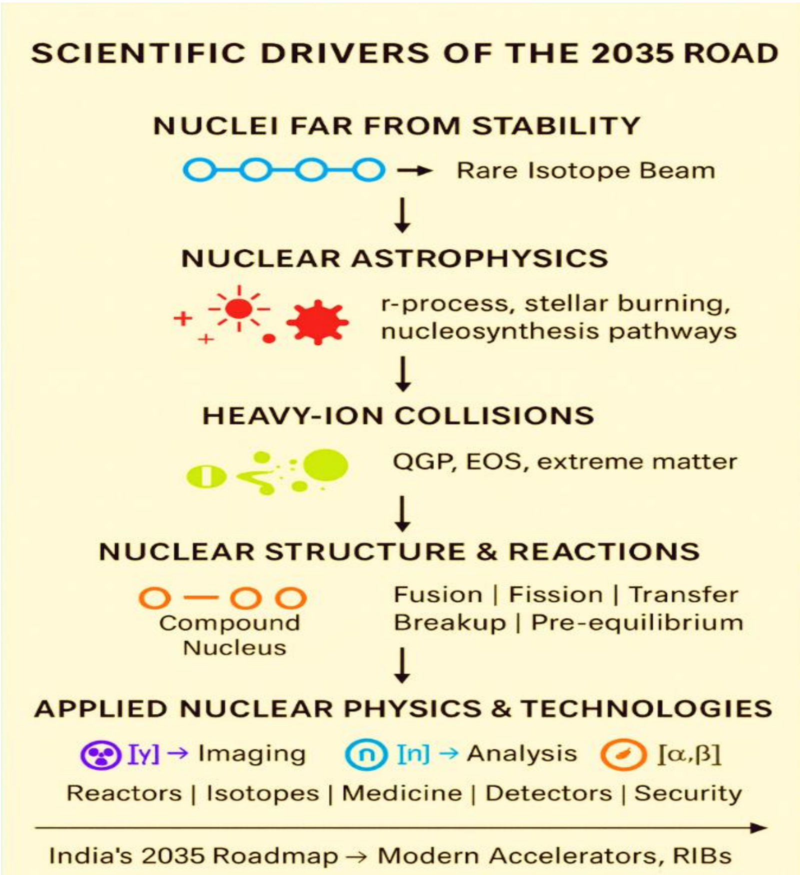
Scientific Drivers of the 2035 Roadmap

The 2035 roadmap is shaped by several scientific frontiers that are reshaping nuclear physics worldwide. One of the major drivers is the need to investigate nuclei far from stability, which enables exploration of shell evolution, exotic decay modes, and the fundamental limits of nuclear existence. Nuclear astrophysics forms another critical pillar, aiming to explain nucleosynthesis pathways, r-process element formation, and the nuclear reactions that power stars and explosive stellar events. High-energy heavy-ion collision experiments provide insight into the properties of quark-gluon plasma, the equation of state of nuclear matter, and extreme densities and temperatures comparable to those of the early universe.

At the same time, nuclear structure and reaction studies, including fusion, fission, breakup, transfer, and pre-equilibrium processes, remain central to understanding nuclear dynamics across a broad energy range. Equally important are the applied branches of nuclear physics, encompassing reactor science, isotope production, radiation applications, detector development, nuclear medicine, and national security technologies. The roadmap aligns India's research priorities with these global advancements, emphasising the need for modern accelerators, precision detectors, rare-isotope beam capabilities, and large-scale computational tools.

Current Landscape and Key Challenges

India's nuclear physics ecosystem is widely distributed across universities, IITs, IISERs, DAE institutions, and national laboratories, forming a diverse network with several notable strengths. The Pelletron–LINAC facility at IUAC, New Delhi, the accelerator complex at VECC Kolkata, and the extensive experimental infrastructure at BARC and TIFR are central pillars of experimental nuclear research. India also has active groups working in nuclear astrophysics, detector development, and theoretical nuclear physics, many of which maintain strong international collaborations. However, the Vision–2035 document emphasizes that despite these strengths, there are significant structural challenges that threaten the long-term growth of the field. A major concern is the shrinking human-resource base: faculty positions in universities have declined, dedicated nuclear physics courses are disappearing, and young Ph.D. graduates face limited career opportunities. Representation of nuclear physicists in national scientific committees has also weakened over time. The document further highlights ageing or insufficient infrastructure. Many accelerators are operating near saturation; university laboratories lack modern experimental facilities and high-performance computing (HPC) clusters remain inadequate for advanced theoretical work. India still does not have a national underground laboratory for low-background experiments such as dark-matter searches or double-beta decay. In addition to these issues, the absence of a unified, nationwide coordination mechanism has historically fragmented efforts, limited outreach, and hindered systematic planning, an issue now addressed by the establishment of INPA.



Infrastructure Roadmap: 2025–2035

The infrastructure roadmap proposed in Vision–2035 outlines a phased and strategic expansion of India's experimental and theoretical capacities. Upgrading existing facilities is a core priority. The IUAC Pelletron–LINAC system requires enhancements in beam intensity, new-generation detector arrays, and improved RF systems to support precision studies in structure and reaction dynamics. Similar modernisation is recommended for BARC and TIFR facilities, which need advanced detector systems and refined accelerator components to expand research in nuclear reactions, spectroscopy, and materials analysis. The VECC

Kolkata is expected to strengthen its cyclotron capabilities and advance radioactive ion beam (RIB) development to support rare-isotope and astrophysics research. Beyond strengthening existing laboratories, the roadmap calls for the establishment of several new national facilities. High-performance computing centres dedicated to nuclear structure, reaction modelling, lattice QCD, and astrophysical simulations are crucial for theoretical advancement and for meaningful participation in global collaborations. The document also stresses the need for dedicated detector development centres capable of indigenous fabrication of silicon, scintillator, neutron, gaseous, and cryogenic detectors, which are critical for scientific self-reliance and potential export capability. A final but essential component of the infrastructure plan is the creation of regional university laboratories equipped with small accelerators, neutron sources, and gamma- and charged-particle experimental setups. These centres would serve as training hubs, ensuring equitable access to laboratory facilities for students across the country.

Human Resource and Training Strategy

Strengthening human resources is at the heart of the Vision–2035 roadmap, which identifies manpower development as the most urgent need for sustaining India's nuclear physics program. The revival of national training schools is a top priority, including annual schools on nuclear physics, detector technologies, and computational methods. These schools would provide structured training for students, early-career researchers, and faculty.

Equally important is the creation of internship and fellowship opportunities that allow students to work at national laboratories during summer and winter breaks. Joint doctoral supervision across universities and national labs, along with doctoral exchange programs, is encouraged to diversify training and enhance research capabilities. The roadmap proposes the creation of a national database of nuclear physicists, maintained by INPA, to catalogue researchers, facilities, projects, publications, and training opportunities. Such a database will serve as an essential tool for coordination, planning, and visibility. The roadmap strongly advocates empowering young researchers through mentorship networks, early-career research grants, travel support, and representation on scientific committees. This ensures their integration into national decision-making processes and provides a clear pathway for career progression.

Role of INPA in Achieving Vision–2035

The establishment of the *Indian Nuclear Physics Association* represents a major step toward unifying the community and transforming the Vision–2035 roadmap into measurable outcomes. The INPA is envisioned as a representative body that will advocate for modern infrastructure, engage with the PSA's office, DAE, DST, SERB, and UGC, and ensure fair academic representation for nuclear physicists across national policy forums. INPA will play a central role in strengthening education and training by coordinating schools, workshops, and outreach programs. It will maintain the national database of nuclear physicists, thereby enabling effective planning and collaboration. INPA also aims to foster international partnerships, promote public engagement, and serve as the central communication channel through the *Nuclear Horizons* bulletin. By mobilising collective expertise across institutions and generations, INPA provides the mechanism through which the nuclear physics community can convert national vision into sustained action.

Global Benchmarking and Learning

The Vision–2035 roadmap acknowledges the importance of learning from international models of scientific organisation and growth. Countries such as China, Japan, Germany, and the USA have demonstrated that long-term strategic planning, large-scale coordinated investments, and sustained government support can transform national nuclear physics ecosystems within a decade. China's rapid rise, fuelled by integrated planning, advanced accelerator development, and strong community organisation, serves as a compelling example of what collective effort can achieve. By benchmarking against global facilities and adopting best practices, India can identify infrastructural gaps, prioritise investments, and position itself effectively in the international nuclear science landscape.

Strategic Importance for India

Nuclear physics is of profound strategic relevance to India, extending far beyond academic research into critical national sectors. A robust nuclear physics program is essential for nuclear energy security, particularly for future reactor designs, fuel-cycle innovations, and advanced waste-management solutions. Medical isotope production and nuclear medicine rely heavily on nuclear reaction data, accelerator technology, and detector development. Space science missions increasingly use nuclear detectors and radiation modelling tools, while national security applications depend on nuclear forensics, radiation detection, and non-proliferation technologies. Materials research, radiation applications in agriculture, and energy studies further underscore the pervasive utility of nuclear physics. Given these broad contributions, strengthening nuclear physics is not optional, rather it is a strategic necessity that underpins scientific sovereignty, technological leadership, and national preparedness.

Roadmap Summary: Milestones (2025–2035)

The Vision–2035 roadmap outlines a clear sequence of milestones that gradually build India's nuclear physics capabilities. By 2027, the priority is to establish a fully functional INPA, develop a national database of nuclear physicists, initiate a large-scale membership drive, revive national training schools, and formalise proposals for a high-performance computing cluster. By 2030, significant progress is expected in modernising detectors and accelerator systems at IUAC, VECC, and BARC, expanding mentorship and national internship programs, and upgrading the first phase of university laboratories. The establishment of a Detector Development Centre marks an important mid-term goal for indigenous technological capability. By 2035, the roadmap envisions the completion of a strong computational ecosystem, and the commissioning of next-generation accelerators and radioactive ion beam facilities. These developments, combined with a sustained pipeline of trained manpower, would position India as a leading centre of nuclear physics research in Asia and a key global contributor.

Conclusion

The ***Mega Science Vision–2035: Nuclear Physics*** document presents a comprehensive and forward-looking blueprint for revitalising India's nuclear physics ecosystem. Through strategic investment in modern infrastructure, a renewed emphasis on manpower development, and strong national coordination, India can build a vibrant and globally competitive nuclear physics program. The creation of the Indian Nuclear Physics Association signals the beginning of a new phase of collective action and community-led growth. With unified effort, transparent governance, and sustained commitment, the nuclear physics community can successfully transform the 2035 roadmap into reality and lay the foundation for an even more ambitious *Nuclear Physics Vision 2050*.

Reference: *MEGA SCIENCE VISION – 2035 (NUCLEAR PHYSICS): A roadmap prepared by the Indian Nuclear Physics Community with TIFR, Mumbai as the Nodal Scientific Institution and submitted to The Office of the Principal Scientific Adviser to the Government of India.*

Prof. B. P. Singh *is Senior Professor and a faculty member in the Department of Physics, Aligarh Muslim University (AMU), Aligarh. His academic and research interests are centred on nuclear reaction dynamics, heavy-ion collisions, pre-equilibrium emission, and the applications of nuclear physics in science and society. He is actively engaged in teaching, research, and scientific outreach, and is committed to promoting nuclear science education through academic publications and public-engagement initiatives.*

Promotion of Nuclear Physics with advanced funding for teaching and research in the University system in the country -A much-needed action plan for Nuclear Physics

S. C. Pancholi and R. K. Bhandari

Abstract: *Emphasis is laid on the much-needed action plan for the upgradation and expansion of infrastructure for teaching and research in nuclear physics in universities across the country. This need is driven by India's pathbreaking nuclear mission to meet our growing energy requirements over the next two decades, as well as the worldwide growth of fundamental nuclear physics and its technological spin-offs in medicine, industry, agriculture, space, and other areas beneficial to humankind. A clear view, therefore, emerges for the immediate need for enhanced funding in this important sub-field of science within universities.*

Introduction

Research in all branches of science is necessary for balanced advancement and development of the country. However, research in nuclear physics occupies a place of fundamental importance, as phenomena in nuclear physics are omnipresent in the Universe and in all walks of life. If you gaze at the night sky, the glitter of the stars is ever exciting and beautiful. The source of energy in the stars and the Sun is nuclear reactions, which one does not realise in everyday life.

One needs to emphasise here that basic teaching and research in nuclear physics generates a robust knowledge base that is critical for the sustainable development of our mega power requirements in the coming two decades and for technological applications for the benefit of mankind. It has led to spin-offs in many high-tech areas, such as supercomputers, reactor technologies, and technologies for security and medical applications.

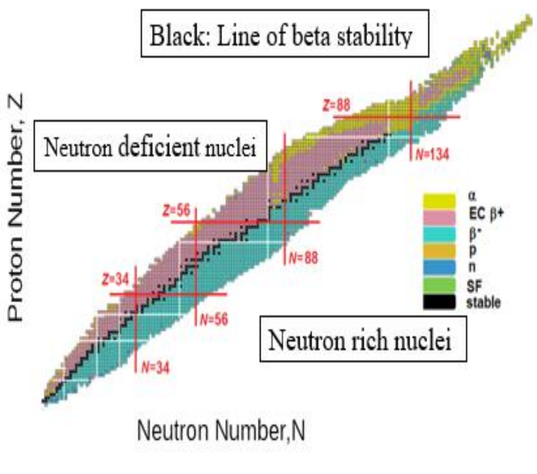
To put more focus on applications of nuclear physics, deeper studies in nuclear sciences in general and nuclear physics in particular are essential for technological developments that will steer the country towards self-sufficiency and self-reliance in several key areas. Nuclear power is an important green power source that needs to be developed using indigenous resources in the near future for climate control, for obvious reasons. In this connection, reference is invited to a historic decision taken earlier in the year 2025 by the Government of India to ramp up the generation of nuclear power to 100 GW (from the existing about 22 GW) by 2047. This will amount to about 10% of the total power generation. Public sector and private sector collaboration would be beneficial to rapidly achieve this goal. A key focus of this mission would also be to develop Small Modular Reactors (SMRs) with generation capacity ranging from 16 MW to 300 MW. With the knowledge and expertise already available in the country, emphasis would be on indigenous R&D. The SMRs are particularly suited for remote areas and energy-consuming industrial clusters. It is also relevant to mention that India has an ambitious three-stage program for this purpose, in the long run, to utilize its abundant thorium reserves.

A large pool of scientists and technologists trained in nuclear physics will be urgently needed in the near future. This expertise is essential for radioisotope production used in medical diagnostics and cancer therapy, helping make these technologies affordable for the masses. Radiation detectors routinely used in low- and high-energy nuclear physics experiments find direct applications in PET and SPECT scanners. Similarly, particle accelerators used for basic nuclear physics research play a crucial role in the economical production of semiconductors.

A strong grounding in nuclear physics is also required for simulating radiation damage in spacecraft electronics, an important aspect of space missions. Nuclear techniques continue to be central to areas such as radiocarbon dating, pollution monitoring, food preservation, and radiation-induced mutation in agriculture. Looking ahead, deep space exploration is another frontier where nuclear technology is expected to play an increasingly significant role. In this article, an attempt will be made to elaborate on a few of the above aspects, to bring home, the paramount need of promotion and expansion of nuclear physics research and teaching with advanced funding in the country.

Global major experimental facilities

At the outset, it is important to highlight the unique experimental facilities available in the country for nuclear physics research. These facilities must be continually expanded and upgraded to keep pace with global advancements. Their uniqueness stems from the fact that, in India, research largely focuses on neutron-deficient nuclei, whereas Western countries and Japan primarily investigate neutron-rich and hard-to-populate nuclei.



Nuclear isotopes of various elements are arranged on the chart of nuclides, with protons on the y-axis and neutrons on the x-axis. Stable nuclei lie along a narrow slanting band called the line of beta stability, with neutron-deficient nuclei on the left and neutron-rich nuclei on the right. With the facilities currently available in the country, we can also explore several neutron-rich nuclei.

The Inter University Accelerator Centre (IUAC) in New Delhi is a centralised facility for research in diverse fields of physics within the university system under the University Grants Commission (UGC), New Delhi. It houses a 15UD tandem Van de Graaff accelerator (Pelletron Accelerator) coupled with a superconducting linear accelerator (LINAC) to boost the energy of the heavy ions accelerated by the Pelletron accelerator. Recently, at this Centre, a High Current Injector (HCI) accelerator is under the commissioning process to further strengthen nuclear physics research. In addition to these, several low-energy accelerators are also operational at this institution. At the Tata Institute of Fundamental Research (TIFR), Mumbai, a Pelletron accelerator similar to the one mentioned above, along with a superconducting linear accelerator, are operational under the BARC–TIFR programme. At the Variable Energy Cyclotron Centre (VECC), Kolkata, a room-temperature cyclotron (K = 130) and a recently commissioned superconducting cyclotron (K = 500) support a wide range of nuclear physics research. A Radioactive Ion Beam Facility is also being developed at VECC. Besides these major centres, several smaller accelerators across the country serve limited research needs in both university and DAE systems.

Internationally competitive research in nuclear structure and nuclear reactions is carried out using the accelerator facilities at IUAC, BARC–TIFR, and VECC. Extensive accounts of these high-quality research activities and experimental capabilities are available in research papers and review articles[1-5]. Additionally, books[6-8] on nuclear structure published in the last decade provide valuable guidance for advanced students and researchers.

Nuclear astrophysics, a rapidly expanding field for nearly a century, continues to witness active experimental and theoretical contributions from several groups in India. The FRENA accelerator facility at the Saha

Institute of Nuclear Physics (SINP), Kolkata, is operational, while IUAC is developing a dedicated nuclear astrophysics setup employing intense beams from the HCI accelerator and a low-background gamma-ray detection system.

Major international accelerator facilities—such as FRIB at Michigan State University, CERN in Geneva, FAIR at GSI (Germany), and RIKEN in Japan—are advancing research on neutron-rich and exotic nuclei using radioactive ion beams and on-line isotope separators. In parallel, VECC has undertaken significant accelerator R&D towards establishing the Advanced National Facility for Unstable and Rare Ion Beams (ANURIB)[9]. For further details, readers may consult the Mega Science Vision–2035 Nuclear Physics document[10].

Some emerging views

In view of the focused national and global thrust on nuclear physics as detailed above, we in India need to pay urgent attention to enhancing broad-based scientific programmes geared towards the expansion and development of infrastructure and manpower in this highly relevant area. Universities remain the primary source of young, talented scientific manpower for research, and there is a growing need to strengthen this. Expanding advanced-level nuclear physics teaching, inducting more specialised faculty, enhancing experimental infrastructure, and substantially increasing research funding are essential steps.

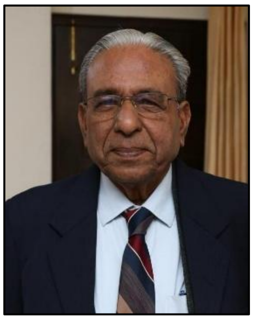
A proven model for supporting high-quality research is the creation of centralised experimental facilities like IUAC which currently supports researchers from over 100 universities and provides an active platform for scientific interaction. Given the size of our country, more Centres like IUAC are urgently needed across different regions to ensure balanced access to research facilities and opportunities in nuclear physics and allied fields. Greater synergy between national laboratories and universities would further accelerate progress; this has already been demonstrated by the Indian National Gamma Array (INGA)[4] collaboration. The study of nuclear phenomena requires a strong integration of experiment and theory, with both streams contributing to a complete understanding. It is therefore important to strengthen nuclear theory research and enhance interaction between theoretical and experimental groups to support future advances in the field.

The authors are highly thankful to many leading scientists in the field for their encouragement to delve into the present research arena, which is much needed for the growth of scientific knowledge in the country. Very useful discussions with them, along with their critical reading and valuable suggestions, have greatly improved the quality of thought and presentation of the manuscript. We are grateful to Professor Avinash C. Pandey for his kind suggestions. The authors further gratefully acknowledge the inputs and suggestions made by Professors N. Panchapakesan, A. K. Jain, and Dr. R. P. Singh.

References:

1. S. Bhattacharya, C. Bhattacharya, P. Das, G. Mukherjee and T.K. Ghosh., **Experimental nuclear physics research using the cyclotron at VECC**, *Eur. Phys. J. A,* 2018, **54**: 158.
2. S. Muralithar, N. Madhavan, P. Sugathan, R. P. Singh, A. Jhingan, R. Kumar, S. Nath, K. S. Golda and J. Gehlot., **Nuclear reaction and structure studies using experimental facilities at Inter-University Accelerator Centre (IUAC),** *Eur. Phys. J. A,* 2022, **58**: 250.
3. Akhil Jhingan, E. T. Subramaniam, S. Venkataramanan, **Detectors and front-end electronics for nuclear physics research at IUAC**, Eur. Phys. J. A (2025) 61:64
4. Rudrajyoti Palit, Piku Dey, Ananya Kundu, **Nuclear structure and reaction studies with digital INGA**, Eur. Phys. J. A (2025) 61:74
5. R. Palit, Md. S. R. Laskar, S. Nag, D. Choudhury, N. Goel, S. Singh and S. N. Mishra, **Review of isomers in the A~135 region and nuclear shape evolution**, *Eur. Phys. J. Spec. Top.*, 2024, **233,** 933.

6. Ashok Kumar Jain, Bhoomika Maheshwari, Alpana Goel, ***Nuclear Isomers-A Primer***, Springer, 2021.
7. S. C. Pancholi, ***Pear-shaped Nuclei,*** World Scientific Publishing Co., 2020.
8. S. C. Pancholi, ***Exotic Nuclear Excitations***, Springer, 2011.
9. Arup Bandyopadhyay, V. Naik, S. D. Choudhury, M. Mondal and A. Chakrabarti, **ANURIB-Advanced National facility for Unstable and Rare Ion Beams***,* Pramana, 2015, **85,** 505.
10. **Mega Science Vision-2035 Nuclear Physics.**, 2023.



Prof. S. C. Pancholi *is a former head of the Department of Physics, Delhi University, Delhi. He has contributed significantly in the development of nuclear structure studies and is a well-known nuclear physicist.*

Prof. R. K. Bhandari *is a distinguished Indian physicist and the former director of the Variable Energy Cyclotron Centre (VECC) in Kolkata. He played a pivotal role in establishing the country's indigenous accelerator capabilities during his more than 40-year career with the Department of Atomic Energy (DAE).*

The Future of Nuclear Astrophysics: Indian Contributions

Akashrup Banerjee
In-Charge, FRENA
Saha Institute of Nuclear Physics, Kolkata

Abstract: *India's contributions to nuclear astrophysics trace their origins to Meghnad Saha's pioneering theoretical work and have evolved into strong experimental capabilities through national accelerator facilities. This progression has culminated in the establishment of the Facility for Research in Experimental Nuclear Astrophysics (FRENA) at the Saha Institute of Nuclear Physics. FRENA provides high-current, well-defined ion beams and supports diverse studies relevant to stellar nucleosynthesis, including gamma spectroscopy, neutron measurements, fast-timing experiments, and charged-particle-gated techniques. Recent detector developments, target innovations, and background-suppression upgrades have enhanced experimental sensitivity. Planned facility upgrades, including advanced spectrometry and national collaborations, position FRENA to play a significant role in shaping the future of nuclear astrophysics in India and globally.*

Current national strengths in nuclear astrophysics

India has long been invested in the field of theoretical astrophysics since the early days of Prof. Meghnad Saha. The Saha ionization equation, introduced in 1920 [1], gives the degree of ionization of a gas in thermal equilibrium and has since been widely used in the field. The seminal papers of Prof. Saha in the early 1920s motivated the Indian scientific community to venture into accelerator-based experiments with the aim of addressing open problems in nuclear astrophysics; a 40-inch cyclotron was set up in Kolkata in the 1950s. Since then, accelerator hubs at New Delhi (IUAC), Mumbai (TIFR), Kolkata (VECC), and Bhubaneshwar (IOP) have led efforts to address astrophysical problems through an experimental nuclear physics approach. Building on the progress of these accelerator centers, a new accelerator was proposed in Kolkata to specifically hunt for answers to open problems in the domain. In this direction, the Saha Institute of Nuclear Physics (Kolkata) established the Facility for Research in Experimental Nuclear Astrophysics (FRENA) in 2018 (Fig. 1). FRENA houses a 3 MV two-stage high-current Cockroft–Walton machine capable of delivering beams of light and heavy ions, with a precise energy definition—critical for astrophysical studies [2]. Over the past three years, several studies have been conducted at FRENA, including accelerator voltage calibration [3] and cross-section measurements, which have put India on the global map of experimental nuclear astrophysics. Over the coming years, FRENA is expected to grow from strength to strength with the support of the vast experience of nuclear physicists in the country, alongside projected advancements in target technology, detector development, data acquisition expertise, and robust theoretical modelling.

Recent experimental advances

With the establishment of FRENA as a national accelerator facility and steadily increasing capabilities to support studies with proton, alpha, carbon, and other heavier beams, a wide variety of experiments are being successfully performed. These include studies of neutron and gamma exit channels of nuclei populated using different projectiles.

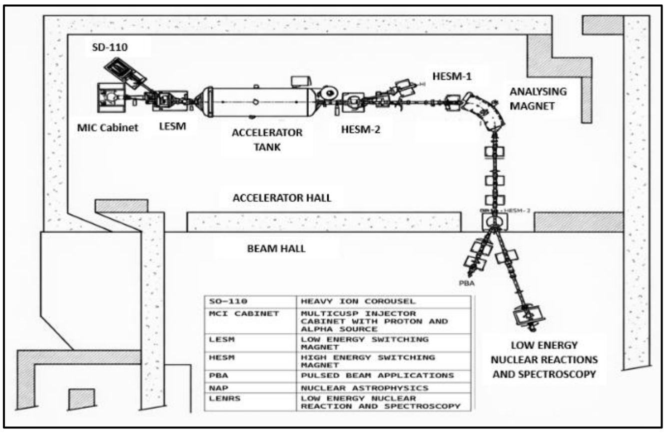


Fig.1(a): *Sketch of the FRENA accelerator, ion source and associated beam lines*

Fig. 1(b): *The accelerator along with the bending magnet and other components.*

Fig.2: *A helical copper ring which is placed 1 cm away from the target and supplied with a cryogenic liquid along and -300 V bias, to create a carbon-free ambience and secondary electron suppression, respectively.*

The facility can support gamma spectroscopy studies with high-resolution solid-state detectors, short-lifetime measurements with scintillating crystals, neutron flux measurements, and charged-particle-gated experiments, among others. FRENA also houses a scattering chamber with a diameter of 1 meter. This enables the study of various stages of the CNO cycle and uniquely supports particle-gated gamma spectroscopy through the provision of placing detectors close to the reaction plane using a gamma cup. In order to sustain targets during high-current measurements, ongoing developments are in place to support target cooling. A helical copper ring with a cryogenic liquid supply has already been fabricated and tested in-house (Fig. 2) to ensure secondary electron suppression as well as to avoid carbon build-up on targets, which is a critical factor for precise reaction cross-section measurements in nuclear astrophysics studies. Furthermore, a new array of BF_3 detectors has been developed by VECC to be utilised for neutron flux measurements at FRENA. An array of $CeBr_3$ detectors for fast-timing measurements has also been commissioned at FRENA.

Facility upgrades relevant to astrophysical reaction studies

In order to stay ahead of international competition in the field of experimental nuclear astrophysics, as well as to effectively compensate for the overground location of FRENA, several upgrades have been planned and are being rapidly undertaken. To identify and measure gamma radiation relevant to nuclear astrophysics, it is imperative to suppress background radiation. It has been planned to use an electronic veto technique to suppress cosmic muon contributions in gamma-spectroscopy studies [4]. A new dual-chamber setup is also being fabricated, which includes two chambers, one in-beam and the other off-beam (Fig. 3).

Fig.3: *Sketch of the dual-chamber gamma-spectroscopy setup at FRENA. The chamber will have provisions for short lifetime studies along with electronic vetoing of cosmic muons.*

Studies involving the measurement of nuclear lifetimes of the order of a few minutes can be performed using this type of setup, in which the irradiation and measurement chambers are located next to each other. The off-beam chamber will be equipped in situ with energy-telescope detectors and particle detectors, and will be surrounded by various gamma-ray measurement devices. The chamber will have SiPM-readout plastic scintillators at the top and bottom for muon detection. The muon data can be vetoed either online or offline using time-stamped data acquisition.

A vision for the next two decades

Since the inception of FRENA in 2008, there have been plans and proposals by SINP faculty members to implement a Recoil Mass Spectrometer to facilitate a wider range of reactions and studies that are crucial for the growth of nuclear astrophysics as a field. Globally, there are only a handful of such facilities that boast both a high-end accelerator and a mass spectrometer. It is envisioned that, in the near future, FRENA will, in collaboration with various national laboratories and universities, prepare a concrete proposal to seek grants in this direction from the Government of India.

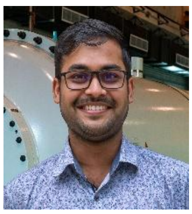
References

[1] M. N. Saha, "Ionization in the Solar Chromosphere", Philosophical Magazine, Series 6, Volume 40, Page 472, 1920.

[2] Akashrup Banerjee and Chinmay Basu, "FRENA: India's first nuclear astrophysics accelerator center", The European Physical Journal Special Topics, Volume 233, Page 2859, 2024.

[3] A. Banerjee, K. Banerjee, et al., "The FRENA accelerator and its beam energy calibration", Nuclear Instruments and Methods in Physics Research Section A, Volume 1072, Page 170179, 2025.

[4] Akashrup Banerjee, "Development of a PERCS active-chamber with novel low background-radiation technology for overground nuclear astrophysics measurements", SERB-SRG funded project, 2023.



Dr. Akashrup Banerjee *is an Associate Professor in the Nuclear Physics Division, Saha Institute of Nuclear Physics (SINP), Kolkata, and is In-charge of the Facility for Research in Experimental Nuclear Astrophysics (FRENA). His research focuses on nuclear instrumentation, accelerator-based nuclear spectroscopy, detector development, and radioactive ion beam experiments*

NUCLEAR HORIZONS: A Bulletin of Indian Nuclear Physics Association

Advances in Nuclear Structure Studies: The Next Frontier

Bhoomika Maheshwari
GANIL, France and VECC, Kolkata
bhoomika.physics@gmail.com

Abstract: *The current transformative phase of nuclear structure physics highlights a critical shift from basic discovery to a focus on precision, production and national capability. The study of complex atomic nucleus continues to reveal new phenomena, with contemporary research heavily focused on exotic nuclear systems featuring extreme neutron-to-proton ratios and long-lived metastable isomeric states. These extreme systems challenge established understanding of nuclear theoretical models while providing crucial insights into fundamental nuclear interactions and astrophysical pathways. Three primary drivers include the next-generation radioactive ion beam facilities, advanced many-body methods for better theoretical predictions, and data-driven solutions using artificial intelligence, machine learning and future potential of quantum computing. Beyond fundamental science, accurate nuclear structure data significantly impacts societal applications such as energy production, medical diagnostics among others. By leveraging the national accelerator facilities, international collaborations and a growing computational ecosystem, India can achieve enduring scientific insight through a deeply integrated approach to experiment, theory and computation alongside a skilled workforce development.*

The atomic nucleus is among the most complex many-body systems known in nature. Governed by the strong nuclear force, interacting protons and neutrons give rise to a rich spectrum of phenomena, such as shell structure, collective motion, deformation, symmetries, and long-lived metastable states. Despite more than a century of study, nuclear structure physics continues to challenge established understanding and reveal new layers of complexity. Nuclear structure research has entered a phase in which precision, prediction, and national capability are as important as discovery. Its significance extends well beyond fundamental science, underpinning applications in energy, medicine, astrophysics, and strategic technologies. In this context, nuclear structure physics occupies a central place in India's long-term science strategy [1] and forms a core element of the national vision for fundamental nuclear research toward 2050.

A defining focus of contemporary nuclear structure studies is the exploration of exotic nuclear systems with extreme neutron-to-proton ratios. These nuclei, often short-lived and far from the region of stability, challenge traditional shell model understanding and reveal how nuclear magic numbers evolve, weaken, or emerge under extreme conditions. Phenomena such as neutron skins, halo structures, and changes in collective motion provide stringent tests of nuclear interactions and many-body dynamics.

The nuclei with N=Z, form a uniquely sensitive class within this landscape. By amplifying proton–neutron correlations, they provide sensitive probes of isospin symmetry, pairing mechanisms, and collective modes. Investigations in these nuclear structure properties also intersect directly with nuclear astrophysics, linking nuclear structure properties to nucleosynthesis pathways and stellar environments.

Long-lived nuclear isomeric states represent another important frontier. These metastable configurations challenge existing theoretical descriptions and require configuration-dependent models that integrate structure, decay dynamics, and angular momentum effects. Beyond their fundamental role in probing nuclear behaviour under extreme conditions, isomers are attracting renewed attention for potential applications in nuclear clocks, controlled energy release, and advanced medical diagnostics. Nuclear structure physics is undergoing a transformative phase driven by three tightly coupled components as shown in Fig. 1: access to nuclei far from stability through radioactive ion beams, advances in predictive theoretical frameworks, and the integration of modern computational and data-driven methodologies. Next-generation

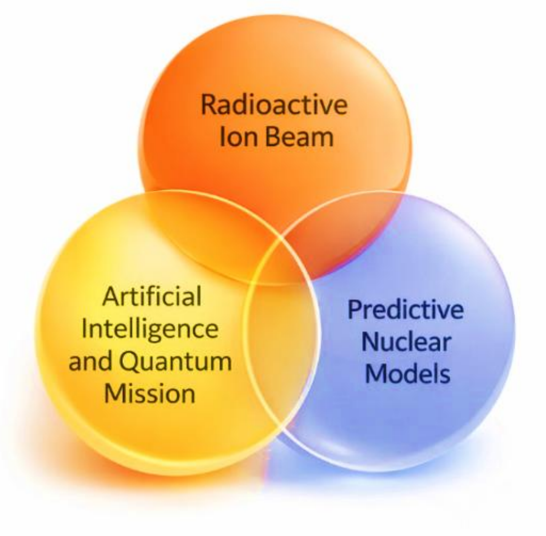


Fig. 1: *Transformative phase components of nuclear structure physics.*

radioactive ion beam facilities have dramatically expanded the experimental frontier. Studies of nuclei with extreme neutron-to-proton ratios provide critical insight into shell evolution, the emergence and erosion of magic numbers, shape coexistence, weak binding, and continuum effects. Facilities such as FRIB and RIKEN are pushing the nuclear landscape toward the drip lines, enabling measurements essential for a unified description of nuclear structure and for constraining astrophysical models of element formation.

In parallel, theoretical nuclear physics has advanced rapidly through high-performance computing and refined many-body methods. Large-scale shell model calculations, energy density functional theory, projected shell models, and ab initio approaches now address systems that were previously beyond reach. Theory increasingly guides experimental strategy and enables controlled extrapolation into unmeasured regions of the nuclear chart, making close integration between theory and experiment central to future progress. Third driver is the growing role of advanced computational techniques, including artificial intelligence and machine learning. These tools are beginning to influence detector calibration, data analysis, model optimization, and uncertainty quantification. Looking ahead, quantum computing offers the potential to address nuclear many-body problems that remain intractable with classical approaches, in line with national initiatives in quantum technologies [2].

Within this global landscape, India has established a robust experimental foundation through accelerator facilities at BARC-TIFR, IUAC, and VECC, supported by national detector arrays. Research from these facilities has contributed significantly to the understanding of nuclear deformation, collective excitations, shell structure, and symmetry-related phenomena, particularly in regions where competing structural effects coexist. Indian nuclear structure research has been further strengthened through sustained engagement with major international facilities, including GANIL, GSI, and CERN. These collaborations provide access to complementary experimental capabilities and enable precision spectroscopy, lifetime measurements, and systematic studies that extend beyond the scope of national facilities alone. Recent DAE symposium proceedings clearly reflect this momentum [3]. India's ANURIB initiative at VECC represents a timely opportunity to align national capabilities with the global thrust toward studies of exotic nuclei.

Advances in nuclear structure research have direct societal impact. Improved nuclear models contribute to the safety and efficiency of nuclear reactors and inform the development of next-generation energy systems [4]. In medicine, nuclear structure data underpin the production of isotopes used in diagnostics and cancer therapy. Applications in nuclear forensics and non-proliferation monitoring similarly rely on accurate and reliable nuclear data. The next frontier in nuclear structure physics lies not only in the discovery of new nuclei, but in the development of a coherent, predictive, and quantitatively reliable framework for nuclear structure. By mid-century, progress will be defined by the ability to make systematic predictions across the nuclear chart with well-characterized uncertainties, grounded in the tight coupling of experiment, theory, and computation.

India is well positioned to contribute strategically by prioritizing selected physics themes and capabilities rather than pursuing incremental breadth. Its established strengths in stable-beam spectroscopy, emerging expertise in studies of exotic nuclei, and a growing computational ecosystem [5] together provide a robust foundation for leadership. Realizing this potential will require a skilled workforce proficient to operate across experiment, computation, and interdisciplinary domains, supported by broader participation and collaborative research environments across universities and laboratories. Sustained emphasis on theory-driven research, advanced data-analysis methodologies, and close integration between experimental and theoretical programs will be essential for long-term impact.

Nuclear structure physics stands at a defining moment. The next frontier is not defined by how far the nuclear chart can be extended, but by how reliably nuclear structure can be understood and predicted. By aligning experimental innovation, theoretical rigor, and computational advances with a long-term strategic vision, India has the opportunity to help shape the global trajectory of nuclear structure research toward 2050 and beyond, transforming data into knowledge and knowledge into enduring scientific insight.

References:

[1] Mega Science Vision – 2035 Nuclear Physics, https://psa.gov.in/CMS/web/sites/default/files/psa_custom_files/MSV2035-NP-Final.pdf
[2] National Quantum Mission, https://dst.gov.in/national-quantum-mission-nqm
[3] Proceedings of DAE Symposium on Nuclear Physics, https://sympnp.org/proceedings/
[4] Nuclear Power in India, https://world-nuclear.org/information-library/country-profiles/countries-g-n/india
[5] Superconducting India 2025, https://supercomputingindia.org

Dr. Bhoomika Maheshwari *is an ANRF Ramanujan Fellow at the Variable Energy Cyclotron Centre (VECC). She was previously an MSCA Research Fellow at GANIL, France, and has held postdoctoral positions at the University of Zagreb. Her research focuses on theoretical nuclear physics, particularly nuclear structure, electromagnetic transitions, and isomers, with recent work exploring machine learning and quantum computing.*

Installation and Commissioning of a Pulsed Neutron Generator Facility at the Department of Physics, BHU, Varanasi

Ajay Kumar Tyagi
Department of Physics, Banaras Hindu University, Varanasi

Nuclear physics activities at Banaras Hindu University began with the early development of its Physics Department in the first half of the twentieth century. In the beginning, the focus was mainly on teaching and building laboratory facilities. As the time progressed, the research activity grew stronger. A major boost came in the late 1960s. During this period, a Van de Graaff generator was installed at BHU. This was one of the early accelerator facilities in Indian universities. It enabled systematic studies of nuclear reactions and particle acceleration. The Department of Physics, BHU went on to train many students in nuclear physics. Several of them later contributed to national laboratories and research centres. Today, nuclear physics research at BHU continues in areas such as nuclear structure, reaction dynamics, and radiation physics. The university remains an important academic centre for nuclear science in India.

Keeping the tradition alive, a significant development took place in March 2022. A D–D reaction–based pulsed neutron generator was successfully installed and commissioned in the Van de Graaff building of the Department of Physics. The machine is developed by the Pulsed Power & Electromagnetics Division, Bhabha Atomic Research Centre (BARC), Visakhapatnam. This facility (neutron generator) is based on plasma focus technology, which is known to produce intense, short bursts of neutrons. It provides a modern platform for advanced research and hands-on training in nuclear science.

A D–D reaction–based pulsed neutron generator works on a simple fusion process involving deuterium (^{2}H), a heavy isotope of hydrogen. When two deuterium nuclei come sufficiently close, they can fuse through the reaction;

$$^{2}_{1}H + \,^{2}_{1}H \rightarrow \,^{3}_{2}He + n + 3.27\ MeV$$

In this process, a neutron with an energy of about 2.45 MeV is emitted along with a helium-3 nucleus, and a total energy of 3.27 MeV is released. In the generator, deuterium gas is introduced and subjected to a very high voltage. This ionizes the gas and creates a plasma of deuterium ions. These ions are accelerated to high energies, and when they collide with other deuterium nuclei, fusion occurs and neutrons are produced.

In pulsed operation mode, the process takes place in very short bursts. Electrical energy is first stored in capacitors and then released suddenly, creating a dense and hot plasma region (often called a plasma pinch). Fusion reactions occur in this compressed plasma, leading to the emission of neutrons in short, intense pulses rather than continuously. Such generators are compact and capable of producing high-intensity neutron bursts with well-defined timing. This makes them very useful for experiments where controlled, time-resolved neutron emission is required.

This particular generator at BHU, is designed to produce a very large number of neutrons, but only for a very short time. In each pulse, it produces $\approx 10^9$ neutrons. Each neutron has an energy of around 2.45 MeV. However, this emission does not last for a long time. The pulse duration is only about 50 nanoseconds, which is an extremely small-time interval. Because of this reason, the neutrons come out in a very sharp and intense burst.

To make this happen, the system first stores electrical energy and then releases it suddenly. The energy is stored in a set of capacitors, which act like electrical storage units by holding charge when a high voltage is applied. In this way, energy is built up in the electric field inside the capacitors. The system can store up to about 10 kJ of energy, which is quite large for such a compact setup. This stored energy is not kept in a single unit but is divided into four separate capacitor modules, each storing about 2.5 kJ. This arrangement helps in better control of the system and also allows the energy to be released very quickly and efficiently when required.

When the system is triggered, all this stored energy is released almost instantly. This produces a very high electric current, reaching up to ≈600 kA. This large current builds up in a very short time, typically about 2 microseconds, and is responsible for creating the conditions required for neutron production inside the plasma focus device. The electrical components of the system are specifically designed to allow fast energy flow. The capacitor bank has very low inductance (about 35 nH) and very low resistance (about 8 mΩ). Low inductance helps the current rise quickly, while low resistance ensures that very little energy is lost in the form of heat. As such, most of the stored energy is effectively used in the process.

The stored energy from the capacitor modules is transferred to the plasma focus device through four switches working together in parallel, known as pseudo-spark switches. These switches can handle very high currents and at the same time allows the energy to be delivered in a controlled and rapid manner. Even though the system handles such high energy and current, it is surprisingly quite compact. The entire setup is mounted on a movable trolley with dimensions of about 1.2 m × 0.9 m × 0.9 m. The total weight of the system is around 300 kg. This makes the entire system portable within a laboratory environment, if needed.

To understand the working in a little more detail, it may be noted that plasma focusing takes place when a strong discharge current flows through a coaxial electrode system inside a chamber filled with deuterium gas. Sometimes it is a deuterium–tritium mixture as well. When the current passes, it creates a plasma that is rapidly compressed into a very small region. This region is called the plasma focus. Here, the temperature and density become high enough for neutron-producing reactions to take place efficiently.

The system is powered by a modular capacitor bank with a total capacitance of ≈50 μF and a stored energy of nearly 10 kJ. This capacitor bank consists of four identical modules, each storing about 2.5 kJ (12.5 μF, 20 kV). The capacitors are designed to have very low inductance (less than about 40 nH), which helps in delivering energy quickly. Each capacitor can handle current reversal up to about **80%** and may carry a peak current of around 150 kA.

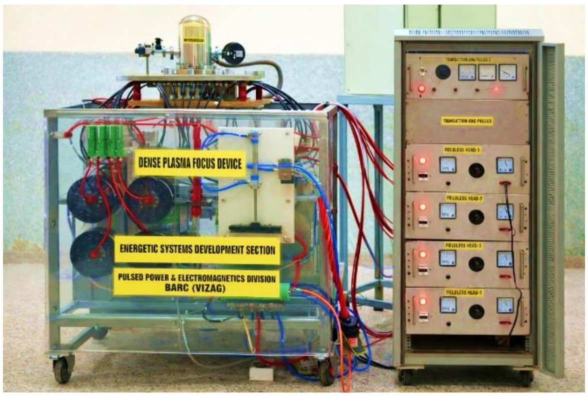


Fig. 1: *A photograph of the newly installed pulsed neutron generator at BHU.*

When all four modules are discharged together, they deliver a combined peak current in the range of about 400 kA to 600 kA. This peak current depends on the charging voltage (typically between 14 kV and 18 kV). The current rises very rapidly, within ≈2 μs, and creates the necessary conditions required for plasma compression and neutron generation.

In a newly established facility, like that at BHU, the neutron yield measurement is essential to evaluate its performance. This also ensures that it is working properly. Further, it confirms the neutron output, repeatability of neutron generation, and also the optimization of the operating conditions for preciseness in experiments.

The neutron yield is measured using a silver activation counter placed at a distance of about 1 meter in the side-on direction. The detection system consists of a moderator, three silver-foil–wrapped Geiger–Müller tubes, a high-voltage power supply, a counter, and a linear amplifier. This setup is well suited for measuring neutron flux in pulsed operation.

The presently installed neutron generator shows optimum performance in single-shot mode at a discharge voltage of about 14 kV and a deuterium gas pressure of around 5 mbar. To maintain consistent neutron output and reduce contamination effects, the filling gas is refreshed after every shot. Under these conditions, an average neutron yield of about 10^9 neutrons per shot is obtained. This makes this machine a special one. The main technical specifications of the neutron generator are summarized below:

Table 1: *Major technical specifications of the neutron generator.*

DD neutron yield per pulse	$\geq 1\times10^9$ n/p (average)
Neutron energy (DD)	~2.45 MeV (typical)
Neutron pulse duration (FWHM)	~50 ns (typical)
Minimum time interval between shots	10 minutes (minimum)
Total capacitance	50 µF (12.5 µF × 4 Nos.)
Capacitor charging voltage	20 kV (maximum)
Operating voltage	17 kV (maximum)
Energy transfer switch	Pseudo spark Switch (× 4 Nos.)
Peak discharge current	600 kA (maximum) @ 17 kV
Overall weight of system	300 kgs (approx.)
Overall size of the system	1.2 m x 0.9 m x 0.9 m

Presently, the neutron generator at Nuclear Physics Lab is being used as a research and training facility by research scholars and M.Sc. students of Banaras Hindu University. Students and researchers from other universities are also using this facility. It is important to note that it is usually operated in single-shot mode, where controlled neutron pulses are produced.

In a typical experiment, samples are prepared and exposed to neutron radiation. After irradiation, the samples are studied using suitable techniques. This helps in understanding the structural, morphological, optical, and electrical properties of different materials. Several experiments have been successfully carried out at the neutron facility in collaboration with University of Rajasthan. These joint efforts have helped in effective use of the facility and strengthened research activities in the area of neutron-based studies.

The facility is also very useful for training. Students learn how to handle nuclear instruments, perform neutron measurements, and analyse the data. It is truly a multidisciplinary platform, where students learn the use of nuclear techniques in different areas such as materials science, radiation studies, and applied physics. This exposure helps them understand real applications and prepares them for work in research laboratories and related scientific fields.

The successful installation of this facility was made possible through the dedicated efforts of many people. The work was carried out under able leadership of the author, with active support from scientists of BARC, Visakhapatnam. The Department of Physics, BHU sincerely acknowledges the strong encouragement and support of A. K. Mohanty, Chairman, Department of Atomic Energy, Government of India. Members of the department are grateful to Dr Archana Sharma and Dr. Rishi Verma from BARC, Visakhapatnam for their constant guidance and support throughout the process.

On behalf of the nuclear physics fraternity, we thank to the Vice-Chancellor of Banaras Hindu University for continuous support and encouragement. Thanks, are also due to the Head of the Department of Physics for all the support provided. Our sincere thanks are due to the technical staff of the department for their dedicated assistance during installation and operation. Last but not the least this work would not have been possible without the help and support of our Nuclear Physics research scholar team members. Their motivation and efforts were vital at every stage.

With the induction of this facility, experimental research activities and the in-house research culture have been rejuvenated. This is truly the need of the hour. It will go a long way in strengthening nuclear physics research and training at the university.

Dr. Ajay Kumar Tyagi *is a Professor and Shri Sant Sanganeria Chair Professor in the Department of Physics at Banaras Hindu University. He is a recipient of the BHU Vice-Chancellor's Award. He has rich international research experience, having worked as a Research Scientist at the Cyclotron Institute, Texas A&M University, USA, and as a Postdoctoral Fellow at the University of Kentucky, USA.*

Inter-University Accelerator Centre: Capabilities and Prospects

S. Nath

Inter-University Accelerator Centre, Aruna Asaf Ali Marg, New Delhi 110067, India

Abstract: *The Inter-University Accelerator Centre (IUAC), New Delhi, is India's premier national user facility for accelerator-based research in nuclear and allied sciences. Established in 1984 and operational since 1991, IUAC provides state-of-the-art accelerator infrastructure to universities and national laboratories, fostering collaborative and internationally competitive research. Its core facilities include the 15UD Pelletron Tandem Accelerator coupled with a superconducting linac, specialized accelerators, advanced detector arrays, and support laboratories. Research spans nuclear structure and reaction dynamics, materials science, atomic and molecular physics, radiation biology, accelerator mass spectrometry, and geochronology. Recent augmentation with the PARAM Rudra supercomputer has strengthened computational and AI-driven research. With upcoming upgrades such as the High Current Injector and the Delhi Light Source FEL, IUAC is poised to remain a pivotal hub for multidisciplinary accelerator science in India.*

Background and Operational Model

The genesis of IUAC traces back to the early 1980s, when the Indian university community articulated a need for a central facility to support accelerator-based research in nuclear and allied sciences. In response, the University Grants Commission (UGC), with approval from the Planning Commission and the Office of the Prime Minister of India, established the Nuclear Science Centre (NSC) in October 1984 as the first Inter-University Centre. Civil construction began in December 1986 on a 25-acre site allotted by Jawaharlal Nehru University in New Delhi.

INTER UNIVERSITY
ACCELERATOR CENTRE

By December 19, 1990, the primary facilities had been set up, and the Centre became operational in July 1991. Reflecting its broadened mission and growing role as a national laboratory, NSC was rechristened the Inter-University Accelerator Centre (IUAC) in 2005. The Centre is funded by the UGC and operated as an autonomous research institution to serve universities, other teaching institutes, and national laboratories across India.

The fundamental objective of IUAC, since its inception, has been to provide front-ranking accelerator-based research facilities to the Indian scientific community, enabling internationally competitive research in nuclear physics, materials science, atomic physics, radiation biology, accelerator mass spectrometry, and allied fields. IUAC operates as a national user facility, where students and researchers can submit experimental proposals. These proposals are vetted by a group of experts for scientific merit and feasibility. Successful proposals are allocated beam time on the Centre's accelerators and access to associated experimental facilities. The operational model of IUAC ensures shared utilization of major accelerator infrastructure and fosters collaboration among researchers from multiple institutions. The Centre plays a significant role in human capital development through the Summer Training Programme for B.Sc. students, the Orientation Programme for M.Sc. students, and the Ph.D. Training Programme. IUAC also conducts specialized schools and workshops for students and early-career researchers to acquaint them with accelerators, associated instrumentation, and accelerator-based research in focused areas.

Research Infrastructure and Domains

The accelerators at IUAC form the heart of its research infrastructure. At the core of IUAC's capabilities is the 15UD Pelletron Tandem Accelerator, a high-voltage electrostatic accelerator capable of delivering stable beams of a wide range of ion species. The Pelletron can hold a terminal potential of up to 15 MV and can deliver both DC and pulsed beams for diverse experiments. To boost beam energies further, the Pelletron is coupled to a superconducting linear accelerator (SC linac). The linac consists of superconducting niobium quarter-wave resonators that accelerate ion beams to higher energies. This Pelletron–linac combination is particularly suitable for nuclear physics experiments. In addition to these two large accelerators, IUAC operates several smaller and specialized accelerators that cater to focused research programmes, besides being useful for the training of users.

IUAC's scientific portfolio spans several interlinked domains of fundamental and applied research. Nuclear physics research is the cornerstone of IUAC's programme. A major thrust of nuclear physics research at IUAC is nuclear structure studies through γ-ray spectroscopy. Such studies began with the Gamma Detector Array (GDA), a modest array of twelve high-purity germanium (HPGe) detectors. The Indian National Gamma Array (INGA), which can accommodate up to twenty-four HPGe Clover detectors, was set up in the early 2000s. INGA enables precision studies of high-spin phenomena, collective excitations, shape coexistence, triaxiality, and exotic modes such as chirality and octupole correlations. Coincidence measurements with charged particles further enhance sensitivity to complex nuclear configurations.

Nuclear reaction dynamics form another core research area. Facilities such as the Heavy Ion Reaction Analyzer (HIRA) and the HYbrid Recoil mass Analyzer (HYRA) enable clean separation and identification of reaction products, making possible detailed studies of near-barrier fusion dynamics, multi-nucleon transfer reactions, and fusion–fission in the heavy-mass region. The National Array of Neutron Detectors (NAND) is suitable for the study of fusion–fission dynamics in heavy nuclei by enabling precise measurements of neutron multiplicities and characterization of the fission fragments. Elastic and inelastic scattering, transfer and breakup reactions, and incomplete fusion can also be studied by exploiting the possibilities of configuring versatile experimental setups in the General-Purpose Scattering Chamber (GPSC). IUAC provides a low-energy, ultra-pure secondary ion beam of ^{7}Be produced via the in-flight method using HIRA. This facility is one of a kind in the country.

Research in materials science at IUAC leverages energetic ion beams to explore how ions interact with matter and to engineer material properties at the micro- and nanoscale. Energetic ions introduce controlled defects, stimulate phase transformations, and drive systems far from thermodynamic equilibrium, enabling studies of fundamental ion–matter interaction processes. Major research themes include ion irradiation effects, nanostructure synthesis, ion-induced mixing and sputtering, modification of mechanical and electrical properties, and surface/interface engineering. In atomic and molecular physics, IUAC employs accelerator-based techniques to investigate interactions of highly charged ions with target atoms and molecules. Ion beams with well-defined charge states and energies allow precise studies of collision dynamics, electron capture and loss processes, excitation and decay mechanisms, X-ray emission spectroscopy, and lifetime measurements of excited states. Research in radiation biology focuses on understanding the effects of ionizing radiation at the molecular and cellular levels, particularly using heavy ions with high linear energy transfer (LET). Experiments are designed to mimic the biological impact of energetic charged particles encountered in space, medical radiation therapy, and radiological environments.

The accelerator mass spectrometry (AMS) programme enables ultra-sensitive quantification of rare and long-lived radionuclides such as ^{10}Be and ^{26}Al for applications in geochronology, Earth sciences, environmental studies, and archaeology. India's first AMS facility, capable of detecting trace isotopes at extremely low concentrations, was established by upgrading the existing 15UD Pelletron during 2004–05. Using AMS, researchers can determine cosmic-ray exposure ages, erosion and sedimentation rates, climate history from paleoclimate proxies, and timelines of geological events with high precision. The National Geochronology Facility (NGF) at IUAC was established with support from the Ministry of Earth Sciences, Government of India, during 2015–20. Its purpose is to provide Indian Earth science researchers with access to state-of-the-art geochronological and isotopic analysis tools that meet international standards, enabling high-accuracy age determination and isotope geochemistry studies.

The research activities of IUAC are sustained by its support laboratories. Specific groups of scientists and technical personnel work in the fields of high-vacuum technology, thin-film development, beam transport, cryogenics, electronics, control, and automation, among others. Complex experiments in nuclear physics are ably supported by the development of customized radiation detectors, analogue and digital electronic modules, and high-throughput multi-parameter data acquisition systems.

Besides setting up and nurturing state-of-the-art experimental facilities, capacity building in computational infrastructure has also been part of the long-term planning at IUAC. The Centre established a high-performance computing (HPC) cluster around 2009–2010 with a modest capability of about 9 teraflops. This system was upgraded in 2012–2013 to a 61-teraflops MPI cluster supported by approximately 55 TB of parallel storage, with financial aid from the Department of Science and Technology (DST), Government of India. Under the National Supercomputing Mission (NSM), the HPC facility at IUAC recently took a giant leap with the installation of a 3-petaflops supercomputing cluster, called PARAM Rudra. The system was developed and deployed by the Centre for Development of Advanced Computing (C-DAC). On September 26, 2024, the Prime Minister of India formally dedicated the PARAM Rudra supercomputer at IUAC to the nation, signifying its operational readiness and strategic importance to research institutions across India. The HPC facility has also opened the possibility of AI/ML research linked to accelerator science, such as predictive maintenance, beam tuning optimization, and sensor data analysis. GPU-enabled nodes within the PARAM Rudra ecosystem support deep-learning workloads and advanced computational frameworks. The installation of the PARAM Rudra high-performance computing system at IUAC marks a significant milestone in the Centre's evolution from a laboratory focused on accelerator-based experimental research in basic and applied sciences into a computationally empowered scientific hub.

Future Prospects

Looking ahead, IUAC is poised to evolve alongside global developments in accelerator science and accelerator-based basic physics research. The High Current Injector (HCI) at IUAC is a major accelerator upgrade designed to significantly enhance beam intensity and broaden the experimental capabilities of the Centre. The HCI will primarily cater to experiments that require high beam currents, which are essential for studying rare nuclear processes, low-cross-section reactions, and applications demanding intense ion irradiation. The HCI comprises a low-energy injector system based on an Electron Cyclotron Resonance (ECR) ion source, capable of producing high-charge-state heavy-ion beams with excellent stability. These beams are pre-accelerated using a radio-frequency quadrupole (RFQ) and a drift-tube linac (DTL) before being injected into the SC linac. This configuration allows efficient acceleration of a wide range of ion species, from light to heavy ions, with substantially higher intensities compared to the conventional injector. Commissioning of the HCI with its full potential and its integration with the SC linac will undoubtedly provide a much-needed fillip to the nuclear physics community in the country. Keeping pace with the accelerator augmentation programme, the experimental facilities of IUAC are also being upgraded in phases.

The Delhi Light Source (DLS), a Free Electron Laser (FEL) facility at IUAC, is a unique accelerator-based light source designed for producing coherent, tunable electromagnetic radiation in the terahertz region. Unlike conventional lasers, the FEL generates radiation by passing a relativistic electron beam through a periodic magnetic structure known as an undulator, allowing continuous wavelength tunability and high peak power. It is currently based on a photocathode-based RF electron gun to produce the electron beam. The DLS is expected to serve a wide range of scientific programmes. In condensed matter and materials physics, it could be used to investigate lattice dynamics, low-energy excitations, and non-linear optical phenomena. The intense, tunable radiation is particularly suited for spectroscopy of phonons, magnons, and molecular vibrations. In chemistry and biology, the FEL enables studies of molecular conformations, reaction dynamics, and energy transfer processes that are difficult to access using conventional sources. The FEL facility complements the Centre's particle accelerator-based physics programme by extending its research capabilities into photon science and accelerator technology development.

Over four decades, IUAC has grown from a visionary centre for nuclear science into a nationally significant laboratory engaged in accelerator-based multidisciplinary research. Besides following its core mandate of catering primarily to the needs of researchers from Indian universities, IUAC has also ventured into a few developmental projects of national importance. The Centre has forged international collaborations in carefully chosen areas to further its strategic and scientific goals. With state-of-the-art accelerators, advanced experimental facilities, and a collaborative research model, IUAC has empowered a myriad of Indian researchers to undertake frontier research. Its continued emphasis on technology development, enhanced capabilities, and expanding scientific horizons ensures that it will remain a pivotal hub for accelerator-based research nationally as well as globally. In doing so, IUAC has played a transformative role in enabling and sustaining high-quality nuclear science research within the Indian university system.

Dr. Subir Nath *is a Scientist-G at the Inter University Accelerator Centre (IUAC), New Delhi. His research focuses on reaction dynamics and nuclear structure, with extensive experience in experiments using recoil mass separators and accelerator-based nuclear physics facilities.*

BARC Accelerator Developments: A Technical Overview

Satyaranjan Santra
Nuclear Astrophysics Section, Bhabha Atomic Research Centre, Mumbai
Email: ssantra@barc.gov.in

Abstract: *This article presents a brief account of the development of different types of accelerators at BARC. A 5.5 MV Van de Graaff single-ended machine, which was commissioned in the 1960s at BARC and utilized for various applications, was converted into a 6 MV Folded Tandem Ion Accelerator (FOTIA) in 2000, with its major components built indigenously. The 14 UD Pelletron Accelerator, which was commissioned at TIFR in 1988 and has been serving as a major facility for heavy-ion accelerator-based research in India, has been connected to an indigenously developed LINAC booster to enhance the energy of the Pelletron beams since 2002. Several electron accelerators with energies in the range of 0.5–10 MeV, having wide industrial applications, have also been built indigenously. Major upcoming accelerator projects for nuclear physics and nuclear astrophysics include the development of a "Stable and Unstable Beam for Heavy Ion Research (SUBHIR) Facility" and an "Underground Facility for Nuclear Astrophysics."*

1. Introduction:

Bhabha Atomic Research Centre (BARC) has a long tradition of indigenously developing different kinds of accelerators for research as well as applications. These accelerators are either DC or RF types and can deliver accelerated beams of electrons or ions. The ion accelerators developed at BARC include the Folded Tandem Ion Accelerator (FOTIA), a LINAC booster, and the Low Energy High Intensity Proton Accelerator (LEHIPA). Similarly, the electron accelerators developed include RF linacs of 500 keV, 3 MeV, and 10 MeV (3 kW and 5 kW), as well as DC accelerators of 0.5 and 1 MeV. A 30 MeV RF linac for research applications, such as shielding studies and n-ToF experiments, is being designed and developed. For ADS studies, a 100 MeV, 100 kW RF linac system is also proposed.

2. Ion Accelerators

2.1 Folded Tandem Ion Accelerator (FOTIA)

Since 1962, a single-ended Van de Graaff accelerator, commissioned at the Nuclear Physics Division of BARC, had been delivering proton and alpha beams with energies up to 5 and 10 MeV, respectively, for nuclear and atomic physics studies. Although the accelerator was purchased, the experimental beam lines, including the switching magnets and experimental setups, were developed indigenously. Due to the limitation in the variety of ion beams from this facility, it was converted into a double-stage Folded Tandem Ion Accelerator (FOTIA) with a maximum terminal voltage of 6 MV in 2000, which is capable of providing beams with atomic mass numbers up to A = 19 [1–2].

The construction of the FOTIA involved the development of technologies for several important components, such as dipole and quadrupole magnets, a high-voltage generator, an SF_6 gas handling system, a PC-based control system, etc. The machine has been successfully running for the last 25 years and is being used for research in basic nuclear physics as well as applied fields such as material characterization and radiation biology.

2.2 Linac Booster for Pelletron Accelerator

The BARC–TIFR 14 UD Pelletron Accelerator, which was commissioned at TIFR, Mumbai, in 1988, has been serving as a major facility for heavy-ion accelerator-based research in

India. The pressure vessel, beam lines, and experimental facilities were developed indigenously. There are five beam ports through which beams are available for experiments. Beams with energies greater than 1 MeV per nucleon over the entire mass range, and above 5 MeV per nucleon for A < 20, are routinely available for nuclear physics experiments. This limits the range of target–projectile combinations.

Therefore, a superconducting linear accelerator has been developed indigenously [3] to boost the energy of the heavy ions emerging from the Pelletron. It has been designed to provide heavy-ion beams with energies greater than 5 MeV per nucleon up to mass A = 80.

The linac booster consists of seven modules, each having four lead-coated (2 μm) copper quarter-wave resonators operating at liquid helium temperature. The development of the superconducting LINAC is a major milestone in accelerator technology in the country. All critical components of the LINAC booster (except the helium refrigerator) have been designed and developed indigenously.

2.3 Low Energy High Intensity Proton Accelerator (LEHIPA)

The Accelerator Driven Systems (ADS) are of tremendous interest due to their capability to incinerate minor actinides (MA), long-lived fission products (LLFP), and to enable thorium utilization as a nuclear fuel for energy production. India has vast resources of thorium; therefore, our interest in ADS lies in utilizing thorium for future clean nuclear energy production with a minimum amount of nuclear waste generation. The ADS mainly consists of three subsystems: a high-current proton accelerator (1 GeV, tens of mA current), a spallation target, and a subcritical reactor. The development of the high-power accelerator was planned in a phased manner, namely 20 MeV, 200 MeV, and 1 GeV. In 2023, BARC completed the first phase of indigenous development of the Low Energy High Intensity Proton Accelerator (LEHIPA), with a target energy of 20 MeV and a peak current intensity of about 2 mA. The standalone LEHIPA accelerator can be used to produce neutrons for materials research and for the production of PET isotopes for diagnosis [4].

The LEHIPA mainly consists of a 50 keV Electron Cyclotron Resonance Ion Source (ECRIS), a 2.7 m long two-solenoid-based Low Energy Beam Transport (LEBT) line, a 4 m long 3 MeV Radio Frequency Quadrupole (RFQ) accelerator, a 1.2 m long Medium Energy Beam Transport (MEBT) line, and a 12 m long 20 MeV Drift Tube Linac (DTL). The accelerating structures are powered using three 1 MW klystrons operating at 352 MHz.

3. Electron Accelerators

BARC develops and uses various electron accelerators, primarily at its Electron Beam Centre (EBC) in Kharghar, Navi Mumbai, for radiation processing, research, and industrial applications such as food irradiation, medical sterilization, and polymer cross-linking. These facilities feature indigenous designs, including DC accelerators (500 keV and 1 MeV) and RF linear accelerators (linacs) with energies up to 10 MeV. These machines are crucial for applications in agriculture, food security, and healthcare, with the technology having been transferred to Indian industries.

The first indigenous Cockcroft–Walton accelerator of 500 keV was commissioned by BARC at BRIT, Vashi, Navi Mumbai, in 1998. Subsequently, BARC set up a new facility, namely the Electron Beam Centre (EBC) at Kharghar, Navi Mumbai, to meet further developmental objectives related to higher-energy electron accelerators and their respective applications [6]. Broadly, two categories of accelerators have been developed: (i) DC Accelerators: Continuous operation, using Cockcroft–Walton multipliers for high voltage, with models such as 500 keV (at BRIT, Vashi) and (ii) RF Linear Accelerators (Linacs): Pulsed operation, reaching energies of up to 10 MeV.

3.1 DC Accelerators (500 keV, 1 MeV and 3 MeV)

The 500 keV DC accelerator, indigenously designed and developed by the Accelerator and Pulse Power Division (APPD), BARC, has been in operation at BRIT, Vashi, for the last 25 years. A beam power of 3 kW operated continuously for 8 hours was established as early as 2001. The operation of the accelerator at 5 kW beam power was demonstrated in 2018. The 500 keV DC accelerator utilizes a 10-stage balanced Cockcroft–Walton multiplier scheme to generate high voltage. The high-voltage (HV) generator is housed in a pressure vessel of 1.1 m outer diameter (OD) and 1.75 m height, filled with nitrogen gas at a pressure of 6 kg/cm^2.

A 1 MeV DC accelerator, based on the symmetric Cockcroft–Walton principle, has been qualified for a ruggedness check involving 24 hours of continuous operation. Presently, the system is operational at a beam power of 50 kW and has been successfully deployed at Unnao (Uttar Pradesh) for the treatment of tannery wastewater as part of the "National Mission for Cleaning Ganga (NMCG)".

APPD, BARC, has also designed and developed a 3 MeV, 10 mA DC electron beam accelerator based on a parallel-coupled self-capacitance-type multiplier (Dynamitron) at the Electron Beam Centre, Kharghar, Navi Mumbai. This machine has been utilized for the reduction of SO_x and NO_x in simulated flue gases and for the treatment of wastewater to reduce COD (Chemical Oxygen Demand) and BOD (Biochemical Oxygen Demand).

3.2 RF Accelerators

10 MeV, 3 kW LINAC: This LINAC has a vertical configuration and is being utilized in various fields. Among these, several applications—including cables, semiconductor modification, gemstone coloration, polymers, and biostimulators—have achieved industrial-level maturity and are being applied for actual utilization. The ruggedness of the system has been successfully tested by running it round the clock reproducibly without any trip.

A new 10 MeV, 5 kW horizontal linac is planned at EBC to increase capacity for industrial users. This will be a compact, tabletop accelerator with a pre-buncher cavity setup to test locally developed components at higher beam power. The beam transmission achieved through the pre-buncher beamline is 99%. Additionally, an improved thermal design has been incorporated in this linac through a jacketed-type cooling arrangement to remove heat efficiently and operate the system continuously at higher power.

6 MeV Single-Energy and 6/4 MeV Dual-Energy Accelerators for Security: A compact 6 MeV RF electron linac has been designed at EBC, which has successfully demonstrated its utility by generating high-resolution images of large cargo. The unit has been validated for radiography of bulk metallic structures and meets ANSI standards for imaging and IEC 62523 norms for material contrast.

9 MeV LINAC: EBC is developing a 9 MeV RF electron linear accelerator for industrial and research applications, particularly for cargo scanning and material processing. It generates high-energy electrons to produce X-rays or for direct treatment and is similar in design to accelerators used by ECIL for national security. Electrons are generated by an electron gun and injected into the linac's accelerating structure. RF power accelerates the electrons to high energies (9–10 MeV). The beam is then directed to an X-ray target to produce high-energy X-rays for imaging or is used directly for material processing.

4. Proposed Accelerator Facilities for Nuclear Physics and Nuclear Astrophysics

4.1. Stable and Unstable Beam for Heavy Ion Research (SUBHIR) Facility at Vizag

A versatile heavy-ion accelerator facility has been proposed to be built at the accelerator complex at Visakhapatnam (Vizag) for accelerating beams of stable and unstable ions with energies up to 10 MeV/u for basic research and applications. It is designed to deliver radioactive ion beams (RIBs) as well as intense stable ion beams across the periodic table, from hydrogen to uranium, with energies much higher than those achievable with the existing BARC–TIFR Pelletron–LINAC facility.

Two separate ECR ion sources are proposed to inject stable and unstable ions into the heavy-ion accelerator, capable of delivering both stable heavy ions (Z in the range 1 to 92) and radioactive ions (Z ≈ 38 to 60, A ≈ 90 to 170). The short-lived nuclear reaction products, mainly unstable fission fragments produced through proton-induced fission and photo-fission of actinide targets, will serve as the source of radioactive ions. Proton-induced fission can provide neutron-rich (via fission) as well as proton-rich RIBs (using **(p, xn)** reactions), while an electron driver can provide neutron-rich RIBs via photo-fission. The facility will feature a state-of-the-art experimental area equipped with advanced detection systems for the measurement of neutrons, gamma rays, and charged particles, enabling frontline nuclear physics research [7].

4.2 An Underground Facility for Nuclear Astrophysics

Accurate knowledge of thermonuclear reaction rates is a key issue in nuclear astrophysics, as it is essential for understanding energy generation, neutrino production, and element synthesis in stars. Cross-section measurements are primarily hampered by very low counting rates and cosmic-ray background. An underground location is therefore extremely advantageous for such studies. We propose an underground nuclear astrophysics facility to be constructed over a 10-year period, with a rock overburden of at least 1.5 km [8]. An ECR ion source (ECRIS)-based, single-ended 5 MV DC accelerator is proposed to be built in the first 7 years, followed by the development of experimental facilities in the next 3 years, at an estimated cost of ₹300 crore (excluding tunnel construction costs). The experimental programs at this facility will mainly focus on helium- and carbon-burning reactions, which occur at higher temperatures (i.e., higher energies) than hydrogen-burning processes. In particular, the reactions $^{12}C(\alpha, \gamma)^{16}O$, $^{12}C + ^{12}C$, and (α, n) reactions on ^{13}C and ^{22}Ne are of critical importance.

References:

[1] S. Santra and P. Singh, *Pramana – Journal of Physics*, **59**, 53 (2002).

[2] S. Santra, K. Mahata, P. Singh, C. V. Fernandes, M. Hemalatha, and S. Kailas, *Nuclear Instruments and Methods in Physics Research A*, **496**, 44 (2003).

[3] R. G. Pillay, M. B. Kurup, A. Jain, D. Biswas, S. A. Kori, and B. Srinivasan, *Indian Journal of Pure and Applied Physics*, **27**, 671 (1989).

[4] U. D. Malse, *BARC Newsletter*, October 2023, Founder's Day Special Issue, 48 (2023).

[5] K. C. Mittal, "Industrial Electron Accelerators: Development and Applications," Invited Talk, InPAC-2013, Kolkata (2013).

[6] Archana Sharma, *BARC Newsletter*, October 2023, Founder's Day Special Issue, 54 (2023).

[7] A. Shrivastava, *BARC Newsletter*, **399**, 83 (2025).

[8] S. Santra, *BARC Newsletter*, **399**, 80 (2025)

Dr. S. Santra *is the Head of the Nuclear Astrophysics Section, Physics Group. He is a well-recognized expert in nuclear reaction dynamics, with extensive contributions to accelerator-based heavy-ion reaction studies, and has played a significant role in advancing experimental nuclear astrophysics research in India.*

Inside the Nuclear Physics Special Laboratory at AMU: Exploring Experiments and the Beta Particle Spectrum

Mohd. Shuaib and Aquib Siddique
Department of Physics,
Aligarh Muslim University, Aligarh-202002, INDIA

The Special Nuclear Physics Laboratory at the Department of Physics, Aligarh Muslim University, Aligarh, India, is one of the finest facilities for teaching nuclear physics at the postgraduate level. It allows students to learn nuclear and radiation physics through hands-on experiments. The laboratory experiments help in linking classroom theory with practical observation. The laboratory is equipped with various alpha, beta gamma-radioactive sources, radiation detectors, spectroscopy systems, and modern data acquisition tools, providing a complete environment for experimental learning in nuclear physics.

Students perform experiments with alpha, beta, gamma radiations, and X-rays to study radioactive decay, radiation–matter interaction, Compton scattering, X-ray fluorescence, and gamma-ray spectroscopy. Using Geiger–Müller (GM) counters, NaI(Tl) scintillation detectors, semiconductor detectors, and multichannel analysers (MCA), students gain practical skills in counting statistics, detector efficiency, and radiation measurement techniques.

Figure 1: *Panoramic view of the Nuclear Physics Special Laboratory at Aligarh Muslim University.*

The laboratory includes experiments to measure the dead time of GM counters using two source method and verify the inverse square law for gamma radiation. Students determine gamma-ray attenuation coefficients using absorbers such as lead, copper and aluminium. They study X-ray fluorescence and verify Moseley's law to understand the relationship between atomic number and X-ray energies. Using the

Compton scattering setup, they observe wavelength shifts and compare measured and calculated scattered photon energies, which nearly match. Compton scattering data is then analysed to determine the rest mass of electron within ≈4–5% accuracy. Students also analyse alpha particles from ^{241}Am to study their emission patterns and discrete energy spectrum. They also measure internal conversion in $^{137m}_{56}Ba$, which provides insight into how the nucleus releases energy and allows them to test and verify nuclear models.

Experiments on detector efficiency help students understand gamma and beta detection, while range, endpoint energies, and absorption coefficients of beta sources are measured to study beta decay. Additionally, students determine foil thicknesses using alpha transmission and perform gamma-ray spectroscopy to study energy resolution with standard sources like ^{60}Co, ^{133}Ba, and ^{137}Cs. The thin foil thickness experiment uses a small vacuum chamber and a ^{241}Am source. It helps students learn about energy loss and the practical use of semiconductor detectors.

Another beautiful example in the laboratory is the pioneering Rutherford experiment set-up. In a vacuum chamber maintained by a rotary pump, monoenergetic alpha particles strike a thin gold foil. One observes that the counts of scattered particles decrease with increasing scattering angle, exactly as predicted by theory. A careful analysis of such scattering data led, Rutherford, to the discovery of the atomic nucleus. This experiment shows how simple observations can reveal the internal structure of atoms.

Among these, one of the most interesting and instructive experiments is the measurement of β-ray spectra of ^{90}Sr and ^{22}Na using Magnetic deflection and a GM counter. This experiment allows students to determine the most probable as well as endpoint energies of β-particles. Students not only learn the working of a gas-filled Geiger–Müller counter but also understand beta decay and energy spectrum measurement. By analysing the β-particle trajectories in a magnetic field and constructing energy spectra, students gain a clear understanding of the continuous nature of beta decay and energy sharing with neutrinos. The details of this experiment are presented below for completeness.

As background it may be remarked that, the radioactive decay is a natural process in which an unstable atomic nucleus releases excess energy in the form of alpha (α), beta (β), or gamma (γ) radiation. In alpha decay, the nucleus emits an alpha particle, reducing its mass by four units and its atomic number by two. In gamma decay, an excited nucleus releases energy as electromagnetic radiation without changing its identity. Beta decay is different, as it involves a change inside the nucleus itself. In beta-minus (β^-) decay, a neutron changes into a proton and an electron are emitted along with an antineutrino. In beta-plus (β^+) decay, a proton converts into a neutron, releasing a positron and a neutrino. A related process, called electron capture, occurs when a proton-rich nucleus absorbs an inner-shell electron (from the K or L shell), converting a proton into a neutron and emitting a neutrino. Early studies of beta decay revealed an important mystery. While alpha and gamma radiations have fixed (discrete) energies, experiments by Lise Meitner and Otto Hahn (1911) and Jean Danysz (1913) showed that beta particles are emitted with a continuous range

Wolfgang Ernst Pauli

of energies. This surprising result was later confirmed in 1914 by James Chadwick [1] who used a magnetic spectrometer along with Geiger counters to accurately measure the beta spectrum (Fig. 1). These historic experiments eventually led to the proposal and discovery of the neutrino, a fundamental particle essential to understanding beta decay.

The measured energy distribution of beta particles did not agree with the law of conservation of energy. If beta decay involved only the emission of an electron, the electron should have a fixed energy value. However, experiments showed a continuous range of energies, suggesting that some energy was missing. This also caused a problem with conservation of angular momentum, since electron emission alone could not account for the change in nuclear spin. To explain this, Pauli in 1931 proposed the existence of a new particle called the neutrino. During beta decay, the neutrino is emitted along with the beta particle. It carries away the missing energy and momentum. The energy is shared between the electron and the neutrino, which explains the continuous energy spectrum.

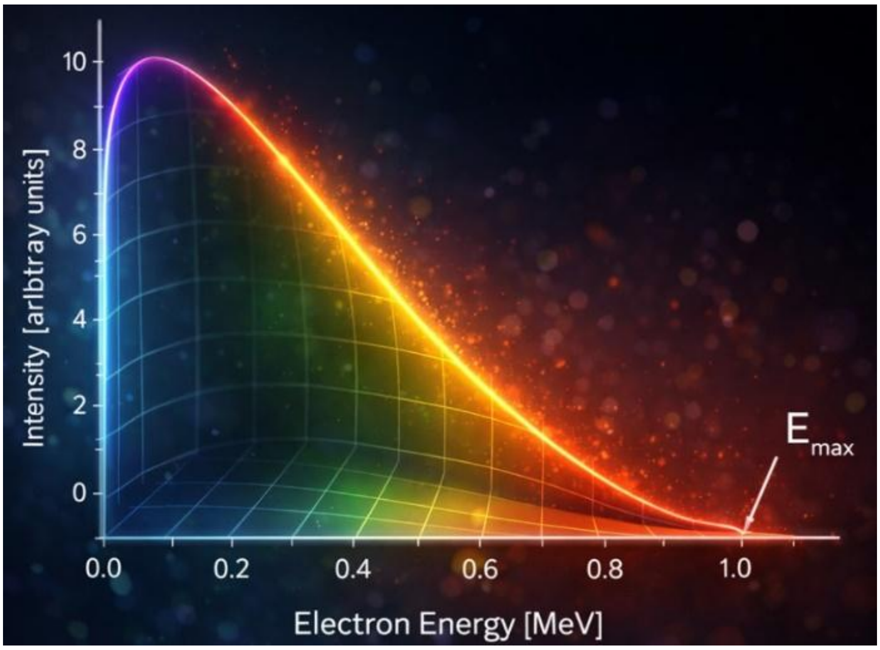


***Fig.2:** A schematic representation of β spectrum.*

In 1934, Fermi theory [2] for beta decay was published, in which the principles of quantum mechanics have been used to describe the beta particle emission. Thus, according to Fermi, electrons are created in the beta-decay process, rather than contained in the nucleus; the same happens to neutrinos. The neutrino interaction with matter was so weak, that its detection is still challenging. Beta decay is mediated by weak force, existence of which was supposed to explain relatively large times and small probabilities of the corresponding reactions. The two types of beta decay are represented as;

The β^- decay is represented by;

$$ {}^{A}_{Z}X \rightarrow {}^{A}_{Z+1}Y + e^- + \bar{\nu}_e \quad (1) $$

However, the positron (β^+) decay is represented by;

$$ {}^{A}_{Z}X \rightarrow {}^{A}_{Z-1}Y + e^+ + \nu_e \quad (2) $$

The β^+ decay or positron decay cannot occur in an isolated proton because it requires energy due to the mass of the neutron being greater than mass of the proton. However, β^+ decay can only happen inside nuclei when the daughter nucleus has a greater binding energy (and therefore a lower total energy) than the parent nucleus. The difference between these energies goes into the reaction of converting a proton into a neutron, positron and a neutrino and into the kinetic energy of these particles. In beta decay, Q is the sum of the kinetic energies of the emitted beta particle, neutrino, and recoiling nucleus. Because of the large mass of the nucleus compared to that of the beta particle and neutrino, the kinetic energy of the recoiling nucleus can generally be neglected. Beta particles can therefore be emitted with any kinetic energy ranging from 0 to E_{max}. Typical value of E_{max} is around 1 MeV, but can range from a few keV to a few MeV.

From equation (1), The Q-value for the β^- decay is given as;

$$Q = \left[m({}^A_Z X) - m\left({}_{Z+1}^{A}Y\right) - m_e - m_{\bar{\nu}_e}\right] \quad (3)$$

Here, $m({}^A_Z X)$denotes the mass of the parent nucleus $({}^A_Z X)$, $m_e$is the mass of the electron, and $m_{\bar{\nu}_e}$is the mass of the antineutrino. The total energy released in the decay equals the mass energy of the initial nucleus minus the combined mass energies of the final nucleus, the emitted electron, and the antineutrino. By neglecting the antineutrino mass and the small difference in electron binding energies, which is negligible for high-Z atoms, we obtain:

$$Q = \left[m({}^A_Z X) - m\left({}_{Z+1}^{A}Y\right)\right]c^2 \quad (4)$$

This is the required energy carried away as kinetic energy by the electron and neutrino. From eq. (5) the reaction will proceed only when the Q-value is positive, i.e., β- decay will takes place when the mass of atom $({}^A_Z X)$ is greater than the mass of atom $({}_{Z+1}^{A}Y)$. Similarly, one can also find the Q-value of β^+ decay as;

$$Q = \left[m({}^A_Z X) - m\left({}_{Z-1}^{A}Y\right) - 2m_e\right]c^2 \quad (5)$$

From eq. (6) one can observe that the reaction will takes place only if Q-value is positive or when the mass of atom $({}^A_Z X)$ exceeds that of ${}_{Z-1}^{A}Y$ by at least twice the mass of the electron.

Generally, it is very difficult to record the spectrum of beta particles emitted in radioactive decay. Here a simple experiment of recording β-particle spectrum using end window β-counter is discussed. The spectrum of two beta sources $^{90}Sr(\beta^-)$ and $^{22}Na(\beta^+)$ has been recorded. Both these sources are beta emitters and their typical decay schemes are given in Fig.3 respectively.

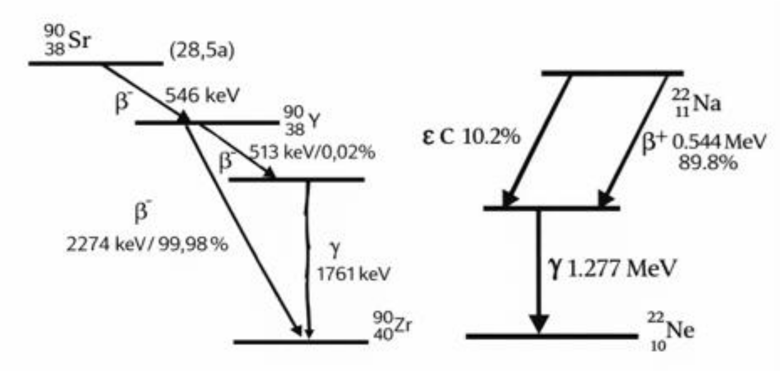


***Fig.3:** The Decay diagram of ^{90}Sr and ^{22}Na.*

The radiation of β-unstable atomic nuclei is selected on the basis of its pulses in a magnetic transverse field, using a diaphragm system i.e., β particle emitted from the two known sources ^{90}Sr and ^{22}Na are selected in the β-spectroscope on the basis of their energy by obliging them to follow fixed orbit using diaphragms in a homogenous magnetic field as shown in Fig. 4. Care is taken while recording the spectrum for ^{22}Na, as it is

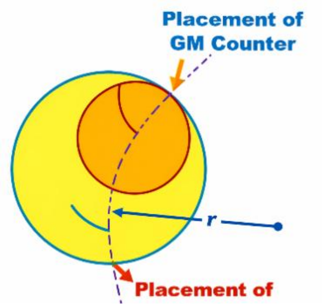


***Fig.4:** Diaphragm system and magnetically deflecte β-particle path and 'r' is the radius of the orbit.*

positron emitter, as such the connections for the coil current are reversed.

In this orbit the Lorentz force, due to the magnetic field and the centrifugal force are in equilibrium i.e.,

$$e.v.B = \frac{mv^2}{r} \quad (6)$$

The above equation yields the following expression for the momentum:

$$P = m.v = e.B.r \quad (7)$$

The equation for relativistic particles with the momentum is given as;

$$\frac{E^2}{c^2} = P^2 + m_0^2c^2 \quad (8)$$

In the above equation E denotes the total energy of the particles, made up of the kinetic energy E_{kin} and m_0c^2 is the rest mass energy of the electron, thus;

$$E = E_{kin} + m_0c^2 \quad (9)$$

From eq. (7), eq. (8) and eq. (9), the kinetic energy can be expressed as;

$$E_{kin} = \sqrt{(eBrc)^2 + m_o^2c^4} - m_oc^2 \quad (10)$$

Now with a give orbital radius of r=50 mm, it is possible to fix a specific particle energy for each magnetic field strength (for each current strength). With the known values of the magnetic field (B), a typical energy calibration has been obtained by using the eq. (10) and is shown in Fig. 5.

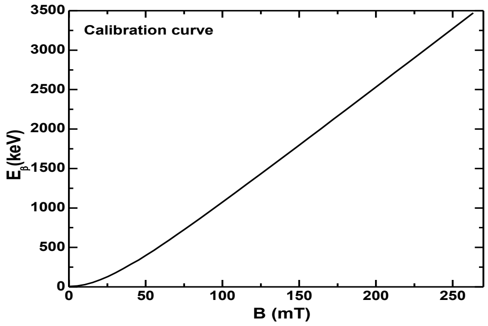


Fig.5: *Calibration curve*

It may be pointed out that from the calibration curve, one can find exactly the energy of the emitted beta particle corresponding to the magnetic field. A photograph of the beta spectroscope system is shown in Fig. 6(A). Here the magnetic field measuring device called Tesla meter, is used along with a tangential Hall probe used into the lateral opening. The source and the counter tube are inserted at respective places in the lateral opening as shown in Fig. 6(B).

Once the setup is ready, the Tesla meter reading should not be disturbed throughout the experiment. By increasing the current of the coil using the power supply, there will be an increase in the magnetic field which is read by the Tesla meter. The counting for the two sources has been done with an interval of 5mT. Consequently, the beta particle of a given energy will be recorded by the end window GM counter. As such one records the current (and hence magnetic field) and the corresponding intensity of beta particle recorded by end window GM counter.

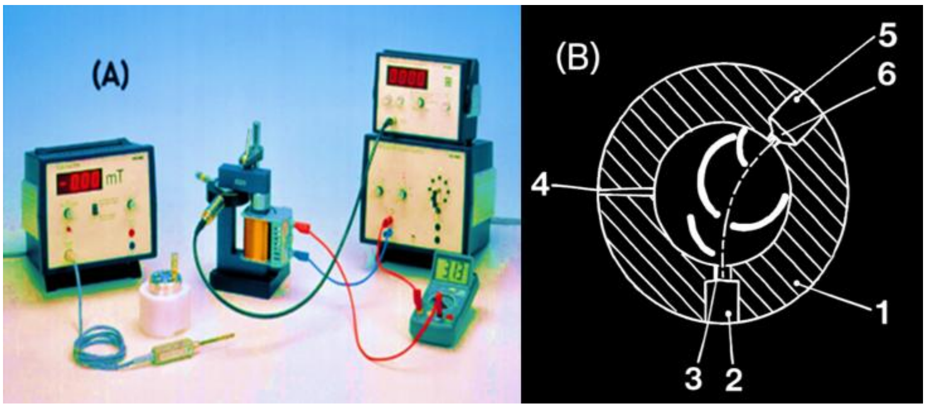


Fig.6(A): *Experimental set up (B): A schematic diagram of β spectrometer: 1: non magnetisable wall; 2: specimen entrance; 3: iris; 4: entrance for tangential Hall probe; 5: counter tube holder; 6: iris.*

Using the calibration curve (Fig. 5), the energy of beta particles is found for each value of the magnetic field. By plotting the number of detected particles against energy, clear beta-energy spectra are obtained for both ^{90}Sr and ^{22}Na, as shown in Fig. 7.

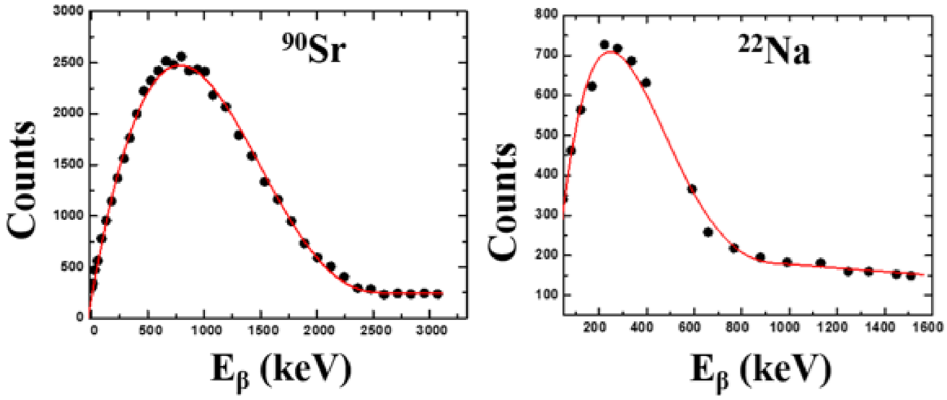


Fig.7: *Experimentally measured beta Spectrums of ^{90}Sr and ^{22}Na*

These spectra clearly show that beta particles are emitted with a range of energies rather than a single value. From these plots, students can determine the most probable energy and the maximum (end-point) energy of the beta particles. The measured values match well with standard reference data, confirming the accuracy of the experiment.

This simple yet powerful experiment helps students understand the nature of beta decay and the role of the neutrino in energy sharing. At the same time, it introduces them to practical skills such as using detectors, magnetic fields, and basic data analysis. Overall, the Nuclear Physics Special Laboratory at AMU offers a hands-on learning experience that strengthens classroom concepts and prepares students for future work in nuclear science, medical physics, radiation technology, and related research fields.

References:

1. G. F. Knoll, Radiation Detection & Measurement, Fourth edition 2010, John Wiley & Sons, USA.

2. R. Prasad, Nuclear Physics, First edition 2014, Pearson, India.

3. Kenneth S. Krane: Introductory Nuclear Physics, 3rd Edition, Wiley, 1987 (ISBN: 978-0471805533).

4. Manual for experiment of PHYWE

Dr. Mohd. Shuaib *is an Assistant Professor at Aligarh Muslim University, Aligarh. He works in the Reaction Dynamics Group and has carried out extensive research on heavy-ion fusion and breakup fusion reaction studies using Pelletron accelerator–based experiments.*

Mr. Aquib Siddique *is a research scholar in the Department of Physics, Aligarh Muslim University, Aligarh. He is working in the field of experimental nuclear physics.*

NUCLEAR HORIZONS: A Bulletin of Indian Nuclear Physics Association

Nuclear Structure Studies at High Angular Momentum

Anindita Karmakar
Saha Institute of Nuclear Physics, Kolkata
(Current Affiliation: Tata Institute of Fundamental Research, Mumbai)
Email: anindita.tifr@gmail.com

Abstract: *The thesis work presents a comprehensive experimental investigation of nuclear structure phenomena at high angular momentum, focusing on the emergence of exotic collective modes such as octupole correlations, γ-softness, and wobbling motion. High-spin states were populated using heavy-ion induced fusion–evaporation reactions and studied through in-beam γ-ray spectroscopy using the Indian National Gamma Array (INGA). Detailed spectroscopic information, including spin-parity assignments, lifetimes, and transition probabilities, provides insight into the interplay between single-particle and collective degrees of freedom in medium-mass nuclei.*

Introduction

Atomic nuclei subjected to high angular momentum exhibit a wide range of structural phenomena arising from the competition between shell effects and collective motion. Away from closed shells, nuclei may develop permanent or dynamic deformations that give rise to rotational bands and vibrational excitations. For the neutron-rich isotopes in mass 100, the level energies of the intruder negative parity $h_{11/2}$ orbital of the neutron sector become significantly lower for $\beta_2 \approx 0.2$. Thus, the shears mechanism becomes a dominant mechanism for the angular momentum generation in these nuclei, where the neutrons occupy the lower levels of the $h_{11/2}$ orbital, while the protons occupy the higher levels of the $(d_{5/2}/g_{7/2})$ orbitals. On the other hand, the occupancy of the $h_{11/2}$ orbital induces nuclear deformation through the polarization of the core and leads to the emergence of several exotic shapes in this mass region. For example, ^{96}Zr (N=56) shows an enhanced electric octupole transition rate B(E3) for the low-lying 3^- state, which is the experimental evidence of octupole instability. This observation indicates that reflection asymmetric characteristics can be present in this mass region. Another recent work on ^{105}Pd reported a transverse wobbling band, which is a signature of a triaxially deformed nucleus. Thus, the search and study of exotic shapes in A ≈ 100 region through discrete γ-ray spectroscopy, is the primary goal for this thesis work.

Experimental Techniques

High-spin states were populated using fusion–evaporation reactions with stable heavy-ion beams at energies of a few MeV per nucleon. The emitted γ-rays were detected using Compton-suppressed HPGe Clover detectors arranged in the INGA geometry. Energy and efficiency calibrations were performed using standard radioactive sources. Spin and parity assignments were obtained from ratios of directional correlations from oriented states (RDCO), and linear polarization analyses. Level lifetimes were extracted using Doppler-shift attenuation methods, allowing the determination of reduced transition probabilities.

Results and Discussion

Detailed spectroscopic studies were carried out for the nuclei ^{100}Ru, ^{98}Mo, ^{105}Pd, and ^{101}Ru. In ^{100}Ru (N = 56, Z = 44), seven interleaved E1 transitions have been observed between the two alternating parity bands whose moments of inertia are nearly identical [1]. The 4 $B(E1)_{out}/B(E2)_{in}$ values for these transitions are nearly constant for both $I^+ \rightarrow (I-1)^-$ and $(I-1)^- \rightarrow (I-2)^+$. There is also a sudden enhancement in this transition rate ratio above spin I = 16 $\hbar$. It is interesting to note that the parity splitting also vanishes in this spin range. The rates along with the vanishing parity splitting indicate a possible emergence of stable octupole deformation in ^{100}Ru in high spins based on a rotationally aligned configuration. From the second experiment with the thick target, the level lifetimes of these bands were estimated. The comparison of the measured B(E1) rates with the calculated values from the Triaxial Projected Shell Model (TPSM), establishes the presence of octupole collectivity at high spins in ^{100}Ru [2]. This is the only nucleus outside the Lanthanide and Actinide regions, where this effect has been observed.

In ^{98}Mo, the spins and multipolarity of the excited levels were established from the present data and few additional transitions were identified. However, due to inadequate statistics, the weak interleaving E1 transitions were not observed [3]. Signatures of dynamic octupole correlations were observed at low spins, highlighting the evolution of reflection-asymmetric structures in this mass region.

The odd-mass nucleus ^{105}Pd shows evidence, for the first time, of the existence of two wobbling bands both having one phonon configuration and originating from the coupling of the wobbling phonon to the ground state band and to its signature partner. The doublet one-phonon wobbling bands are, in turn, found to be the signature partner bands. These observations have been drawn from the measured ratios of the inter-band and intra-band gamma transition rates [3]. The numerical calculations based on the triaxial projected shell model (TPSM) approach have been performed and the obtained results are found to be in good agreement with the experimental observations. These calculations provide an insight into the nature of the observed structures at a microscopic level and establish that the new band originates due to the combined particle-wobbling excitation. This is the first observation of this interplay. The identification of a doublet of wobbling bands establishes the presence of transverse wobbling motion, representing a rare example of this excitation mode outside the well-studied A ≈ 160 region.

Furthermore, studies of ^{101}Ru reveal a transition from wobbling motion to principal-axis rotation with increasing spin, demonstrating the sensitivity of collective modes to changes in angular momentum and quasiparticle alignment.

Conclusion

The present work provides new experimental evidence for the octupole collectivity and triaxial rotation in nuclei at high angular momentum. Through systematic γ-ray spectroscopy and lifetime measurements, the evolution of nuclear shapes and angular momentum generation mechanisms in the A ≈ 100 region has been elucidated. These results offer stringent benchmarks for microscopic models and contribute significantly to the understanding of exotic collective phenomena in finite quantum many-body systems.

References

1. A. Karmakar et al., Phys. Rev. C 109, 054312 (2024).
2. A. Karmakar* et al., Nucl. Phys. A 1066, 123265 (2025).
3. A. Karmakar et al., Phys. Rev. C 110, L051302 (2024)
4. A. Karmakar et. al., Phys. Rev. C 112, 034323 (2025).

Dr. Anindita Karmakar *is a Postdoctoral Fellow at TIFR and was recently awarded the C. V. K. Baba Award for Best PhD Thesis at the DAE–BRNS Symposium on Nuclear Physics held at NIT Jalandhar in December 2025.*

A study of fusion incompleteness in heavy ion interactions at low energies

M. Shariq Asnain[1,*]

[1]*School of Engineering Sciences & Technology, Jamia Hamdard, New Delhi- 110062, INDIA*

* *Department of Physics, Aligarh Muslim University, Aligarh- 202002, INDIA*

Fusion is the phenomenon in which two nuclei combine to create a unified compound nucleus (CN) in an excited state. This excited CN may subsequently undergo decay through the emission of nuclear particles and or gamma rays. The study of heavy-ion (HI) fusion has garnered significant attention from nuclear physicists in recent years. This resurgence of interest can be attributed to the growing accessibility of HI accelerators providing appropriate energy ranges. HI fusion reactions lead to some of the most dramatic reconfiguration that a nuclear many-body system can experience. These reactions allow us to investigate the nuclei at high spins and/or excitation energies. At projectile energies just above the Coulomb barrier (V_B), to well above it, in HI reactions, the most dominating modes of reaction are the complete fusion (CF) and incomplete fusion (ICF) processes. In case of CF, all nucleonic degrees of freedom are involved due to fusion of entire projectile mass carrying input angular momenta $\ell < \ell_{\text{crit}}$. Such fusion leads to the formation of an equilibrated CN. However, in case of ICF reactions, the fusion of the entire projectile with the target nucleus is hindered for the partial waves carrying angular momenta $\ell > \ell_{\text{crit}}$. At higher values of ℓ, the centrifugal repulsive potential increases. As a result, the attractive nuclear potential may not be sufficient to sustain complete fusion of the projectile. To sustain the required angular momentum for fusion, the projectile may break up into fragments. One fragment fuse with the target nucleus, while the remaining part continues forward without interaction.

In this work, the contributions of complete and incomplete fusion in heavy-ion collisions were studied at energies ≈4–7 MeV per nucleon to understand fusion reaction dynamics. Typically, at these energies in HI collisions, the composite system is predominantly formed via the CF process, which is expected to be the sole contributor to the total fusion (TF) cross-section [1]. However, recent observations have indicated a significant ICF contributions [2-4] at these energies sparking renewed interest in the subject under study. Several theoretical models have been proposed to understand ICF reaction dynamics. However, these are found to be reliable at energies ≥ 10 MeV/nucleon, and fail to explain observed ICF reaction data at low energies. Moreover, most of the ICF studies have mostly been carried out using alpha-cluster beams such as ^{12}C, ^{16}O, and ^{20}Ne. While studies using non-alpha-cluster beams like ^{13}C, ^{14}N, ^{18}O, and ^{19}F are still scarce. As such, extending the investigation to study ICF reactions with non-alpha-cluster beams is both interesting and may provide a better insight into the reaction dynamics. In view of this, experimental studies of heavy-ion–induced reactions were carried out for different systems using excitation function measurements for the for ^{14}N+^{181}Ta [5] system and forward recoil range distribution measurements for the ^{19}F+^{159}Tb system [6] system in the energy range ≈4–7 MeV/A. These experiments were performed at the Inter University Accelerator Centre (IUAC), New Delhi, India using the 15UD Pelletron accelerator.

The ^{14}N and ^{19}F beams were focused onto isotopically pure ^{181}Ta and ^{159}Tb, target stacks. To cover the wide range of energy, stacked foil activation technique followed by offline γ-ray spectroscopy has been employed. In the EF measurements, the reaction residues populated via CF and/or ICF processes in ^{14}N+^{181}Ta system have been identified from their characteristic γ-rays and measured half-lives of the residues. As a

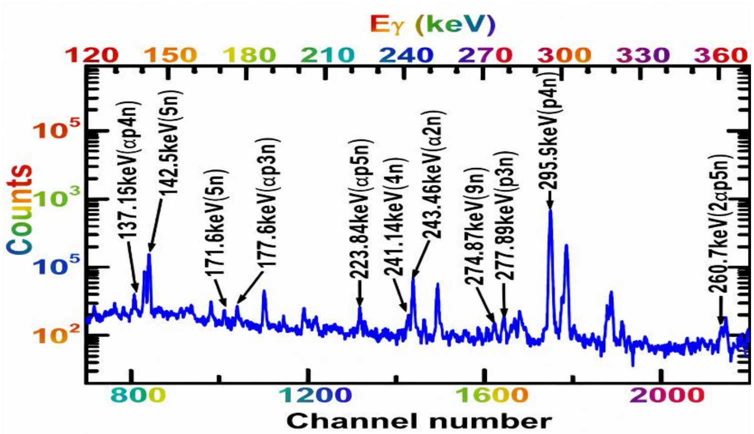


Fig.1: *A typical γ-ray spectrum of $^{14}N+^{181}Ta$ at 87.07 ± 0.93 MeV*

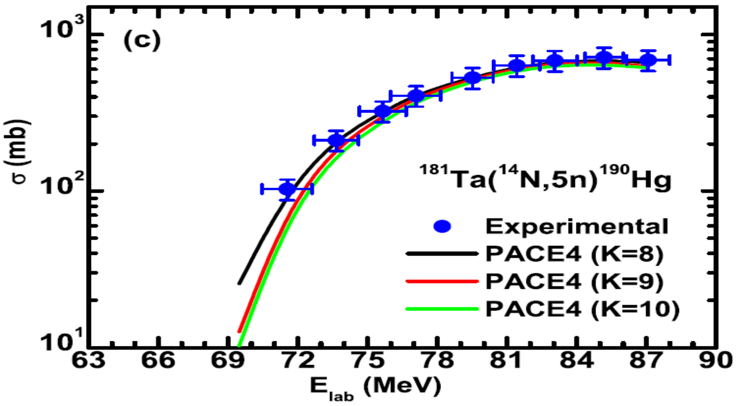


Fig.2: *Experimentally measured and theoretically calculated EFs of ^{190}Hg (5n).*

representative case the relevant portion of the γ -ray spectrum of $^{14}N+^{181}Ta$ recorded at energy87.07±.93 MeV is shown in Fig.1.

The measured cross-section data were analyzed using the statistical model code PACE4 [7]. This code is based on the Hauser–Feshbach theory. It describes how an excited compound nucleus decays. The code includes effects such as nuclear level density and angular momentum. It also considers particle evaporation from the compound nucleus. Standard values for model parameters were used. The calculated results were then compared with the experimental data. The experimentally measured EFs of *xn* and *pxn* channels populated in above system are well reproduced by the PACE4 calculations and confirm the production of these channels solely via CF process. The experimentally measured EF of ^{190}Hg residues populated via *5n* channel is shown in Fig.2, along with PACE4 predictions. It may be pointed out that, in the case of α-emitting channels, the experimental EFs are found to be significantly enhanced with respect to the PACE4 predictions over the entire energy range of study. Since the PACE4 code does not include incomplete fusion (ICF), the observed enhancement in the experimental excitation functions of α-emitting channels can be attributed to ICF processes. Further, the ICF strength function (F_{ICF}), which gives the relative contribution of ICF over CF has also been deduced and its dependence on various entrance channel parameters viz., (i) incident beam energy (ii) Coulomb factor ($Z_P Z_T$) and (iii) α-Q value of the projectile has been studied.

Moreover, an attempt has also been made to understand the role of projectile break-up on fusion cross-section within the framework of Universal Fusion Function (UFF). A significant suppression of about 5–25% in CF cross-section has been observed suggesting that the suppression is due to the prompt break-up of the projectiles. With the help of the experimental fusion function, the suppression factors for different projectiles have also been deduced and a strong correlation between the break-up threshold of the projectile and suppression factor has been observed.

The FRRD measurements provide a clear method to identify incomplete fusion reactions. Using this technique, the relative contributions of complete and incomplete fusion processes were extracted from the degree of linear momentum transfer (LMT) from the projectile to the target nucleus. The ranges of heavy residues, in the stopping medium, populated via CF and ICF processes depend on the recoil velocity, associated with the degree of LMT from projectile to the target nucleus. In CF process, the composite system is formed via full LMT from projectile to the target nucleus. However, in the ICF reactions, the composite system is formed as a result of break up fusion resulting in a partial LMT from projectile to the target nucleus. In the present work, the FRRD of reaction residues populated via CF and/or ICF processes in ^{19}F+^{159}Tb system at two distinct beam energies (83 and 94 MeV) have been measured and reported for the first time [8]. Two different stacks, each consisting of a thin ^{159}Tb target (abundance =100%) followed by a series of thin Al-catcher foils were irradiated separately. The RRD measurements provide us to disentangle CF and ICF events and help to determine their contributions. A significant contribution from ICF reactions was observed, showing their key role in producing these residues. For α-emitting channels associated with incomplete fusion, significant partial and full linear momentum transfer from the projectile to the target is observed. The partial momentum components arise from the breakup of the ^{19}Fprojectile, followed by fusion of fragments such as ^{15}N or ^{11}B with the target nucleus.

As a representative case, the forward recoil range distribution (FRRD) for ^{167}Ybresidues populated through both complete fusion (CF) and incomplete fusion (ICF) processes was experimentally measured at a beam energy of 94.3 MeV, as shown in Fig. 3. By analyzing the measured recoil range distributions, the relative contributions of the CF and ICF components were deduced.

The critical angular momentum, ℓ_{crit}, for the present system was calculated. At this value, the attractive pocket in the entrance-channel potential nearly disappears. The value of ℓ_{crit} was found to be $79\hbar$. The maximum angular momentum, ℓ_{max}, at beam energies of 83 MeV and 94 MeV is approximately $22\hbar$ and $40\hbar$, respectively. Both values are lower than ℓ_{crit}, which allows fusion to occur in the present system. The observation of incomplete fusion at energies where $\ell_{max} < \ell_{crit}$ indicates that a significant number of partial waves below ℓ_{crit} contribute to incomplete fusion reactions.

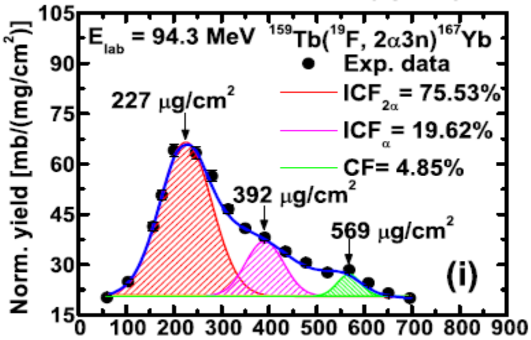


Fig.3: *Experimentally measured FRRD for ^{167}Yb residues populated via both CF and ICF processes at 94.3 MeV*

Further, the validity of Bohr's theory of compound nucleus decay for ^{18}O+^{159}Tb and ^{12}C+^{165}Ho systems, forming the same compound nucleus ^{177}Ta* has also been investigated [9]. In view of the above, the EFs of residues populated in ^{18}O+^{159}Tb and ^{12}C+^{165}Ho systems have been compared with the predictions of statistical model code PACE4. At lower excitation energies, the experimental cross sections for the residues ^{174}Ta(3n) and ^{173}Ta(4n) produced through two distinct entrance channels have been found to be noticeably quite dissimilar as shown by fusion ℓ distribution in Fig.4. However, at relatively higher energies, they coincide for the two entrance channels indicating the validity of independence hypothesis. The observed discrepancy in the cross sections for the two different entrance channels, at lower excitation energies for the residues of interest has been attributed to the mismatch in angular momenta.

The present study indicates that strongly bound non-α-cluster projectiles such as ^{14}N show a significant contribution to incomplete fusion reactions. Loosely bound projectiles like ^{19}F further enhance incomplete fusion yields. These effects persist even at relatively low energies of about 4–7 MeV per nucleon. The results clearly demonstrate that incomplete fusion is a fundamental and universal reaction mechanism at near-barrier energies, extending well beyond the traditional picture limited to weakly bound or α-cluster projectiles. Non-α-cluster systems also play a crucial role in incomplete fusion reaction dynamics.

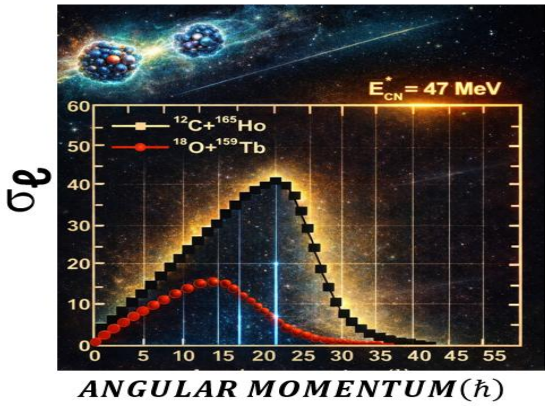


Fig.4: *The fusion ℓ distribution for 18O+159Tb and $^{12}C+^{165}Ho$ systems at 47 MeV excitation energy*

A deeper understanding of incomplete fusion therefore requires additional experimental data using non-α-cluster and exotic beams over a wider mass range. Complementary measurements, such as particle–γ coincidence experiments for different projectile–target combinations, are essential to gain detailed insight into incomplete fusion mechanisms. The resulting data will help refine existing theoretical models and establish universal systematics for heavy-ion reactions at energies around the Coulomb barrier.

References

[1] P. R. S. Gomes et al., Phys. Rev. C **73**, 064606 (2006).
[2] Mohd. Shuaib, et al., Phys. Rev. C **94**, 014613 (2016).
[3] Mohd. Shuaib, et al., J. Phys. G: Nucl. & Part. Phys. **44**, 105108 (2017).
[4] Mohd. Shuaib, et al., Phys. Rev. C **98**, 014605 (2018).
[5] M. Shariq Asnain et al., Phys. Rev. C **104**, 034616 (2021).
[6] M. Shariq Asnain et al., Phys. Rev. C **106**, 064607 (2022).
[7] A. Gavron; Phys. Rev. C **21**, 230 (1980).
[8] M. Shariq Asnain et al., Phys. Rev. C **105**, 014609 (2022).
[9] M. Shariq Asnain et al., Nucl. Phys. A **1041**, 122786 (2024).

Dr. Mohd. Shariq Asnain *was awarded his PhD, in experimental nuclear physics, from Aligarh Muslim University in 2024 and is currently serving as a faculty member at Jamia Hamdard University, New Delhi.*

NUCLEAR HORIZONS: A Bulletin of Indian Nuclear Physics Association

Measurement of Gamma-Ray Attenuation Coefficients in Lead Using a ^{152}Eu Source and Clover HPGe Detector

Lalit Bhardwaj, Aquib Siddique, Mohd. Shuaib, M. Shariq Asnain
Department of Physics, Aligarh Muslim University, Aligarh 202002, INDIA

Abstract: *Gamma rays lose intensity when they pass through matter. This work studies the attenuation of gamma rays in lead. A ^{152}Eu source and a clover HPGe detector were used. Gamma rays with energies from 121 keV to 1408 keV were measured. Lead plates of different thicknesses were placed between the source and the detector. The decrease in gamma-ray intensity was analyzed using the exponential attenuation law. The mass attenuation coefficient of lead was determined for each energy. The results show that attenuation decreases as gamma-ray energy increases. This is due to the change from photoelectric absorption to Compton scattering. The efficiency of the clover detector was also studied. The add-back method was found to improve detection efficiency. The experimentally measured absorption coefficients agree reasonably well with basic radiation theory and help us understand gamma-ray interactions with matter.*

Introduction

Radiation loses its intensity when it passes through matter. This reduction in intensity is called as the attenuation. Different types of radiation behave differently inside matter. Heavy charged particles, such as alpha particles, lose their energy slowly. They undergo many small interactions with atoms of the material. These interactions reduce their energy step by step. The direction of motion usually does not change much. That is why the range of all the charged particles having same energy is not definite that is referred to as the straggling phenomenon.

Gamma rays behave in a very different way. Gamma rays are uncharged and have no mass. When gamma rays pass through an absorber, each gamma-ray photon has only two possibilities. It may pass through the material without any interaction. Or it may interact once and be completely removed from the beam. There is no gradual loss of energy for an individual gamma-ray photon. A simple real-life analogy can help to understand this behaviour. Imagine throwing a large number of identical balls through a forest. Some balls pass straight through without touching any tree. Some balls hit a tree and stop or change direction completely. A ball never loses a little energy at many trees. It either passes through or gets stopped at one tree. Gamma rays behave in a similar way when they pass through matter. Gamma rays can interact with matter through several different processes. However, most of these processes occur very rarely and do not contribute much in practice. In nuclear radiation measurements, only three interaction processes are important because the detection of gamma rays mainly depends on these interactions. These processes are photoelectric absorption, Compton scattering, and pair production.

The chance that a gamma-ray photon passes through matter without interaction is described using probability. Let the thickness of the absorber be x. The probability that a photon does not undergo Compton scattering is $e^{-\sigma x}$, where $\sigma$is the Compton scattering coefficient of the material. The probability that the photon does not undergo photoelectric absorption is $e^{-\tau x}$, where τ is the photoelectric absorption coefficient. The probability that the photon does not undergo pair production is $e^{-\kappa x}$, where κ is the pair production coefficient. Since these three processes act independently, the total probability that a gamma-ray photon passes through the absorber without any interaction is the product of all three probabilities. Therefore, if a narrow and well-collimated gamma-ray beam has an initial intensity I_0, then after passing through an absorber of thickness x, the transmitted intensity I of un-scattered gamma rays is given by;

$$I = I_0 e^{-(\sigma+\tau+\kappa)x}$$

Here, I_0 is the initial intensity of the incident gamma-ray beam, I is the intensity after passing through the absorber, and $\mu = \sigma + \tau + \kappa$, where μ is called the linear attenuation coefficient of the absorber material.

The attenuation coefficient is a measure of the number of primary photons which have interactions with matter. The linear attenuation coefficient can be measured experimentally. However, care must be taken in its use because of its dependence on the density (ρ) of the material for attenuating the gamma rays. Therefore, one generally uses mass attenuation coefficients (μ_o/ρ), a more fundamental value because it is independent of the actual density and physical state (gas, liquid, or solid) of the absorber. In the present work, an attempt has been made to determine the efficiency of the clover detector setup. The gamma-ray attenuation coefficient has also been measured. The measurements were carried out over a broad range of gamma-ray energies.

Experimental Methodology

In the present study, an experimental setup was used to measure the gamma-ray attenuation coefficients in lead. The setup consisted of three main components: a ^{152}Eu gamma-ray source, a clover HPGe detector for measuring gamma-ray intensity, and lead absorbers of different thicknesses to study attenuation. The radioactive source used in this experiment was ^{152}Eu, which has a half-life of 13.33 years. The date of manufacturing of source is 1st May 1983, with an initial activity of 474.63 kBq.

The ^{152}Eu nucleus decays through both β^+ and β^- decay modes. About 72.08% of the nuclei decay by β^+ emission to excited states of ^{152}Sm, while 27.92% decay by β^- emission to excited states of ^{152}Gd. These excited daughter nuclei then de-excite by emitting gamma rays of fixed energies. In the present experiment, several gamma-ray energies were used, covering a wide energy range. The prominent gamma-ray energies are 121.77 keV (28.46%), 244.69 keV (7.51%), 344.28 keV (26.65%), 411.12 keV (2.24%), 443.89 keV (2.81%), 778.92 keV (12.98%), 867.38 keV (4.22%), 964.11 keV (14.53%), 1085.88 keV (9.94%), 1089.70 keV (1.71%), 1112.07 keV (13.63%), 1212.93 keV (1.40%), 1299.15 keV (1.63%), and 1408.00 keV (20.84%). The values given in brackets represent the branching ratios, expressed as percentages.

As already mentioned, a clover HPGe detector was used for gamma-ray detection in this work. This detector is a composite system consisting of four high-purity germanium crystals housed in a single cryostat. The total photopeak efficiency of the clover detector arises from two effects. The first is the complete absorption of gamma rays within a single crystal. The second is the full energy absorption shared between two or more crystals. These combined effects increase the overall detection efficiency of the system. These two effects are shown in Fig. 1, the first effect occur in the crystal 3, and the second effect occurs in the crystal 1 and 2.

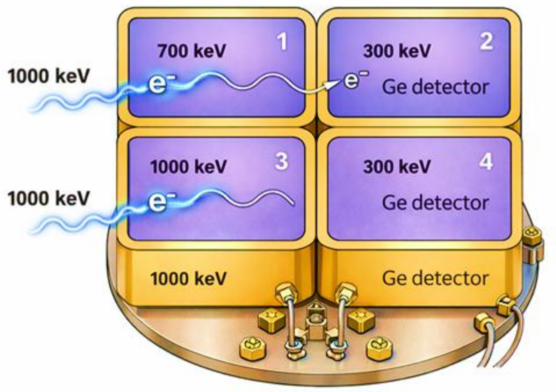


Figure 1: *A schematic representation of two complementary effects in the Clover HPGe detector.*

The former is related to the direct photopeak efficiency $(\varepsilon_P\omega)_D$ and the latter to the add-back photopeak efficiency $(\varepsilon_p\omega)_{AB}$. The total detection efficiency can be written as

$$(\varepsilon_p\omega) = (\varepsilon_p\omega)_D + (\varepsilon_p\omega)_{AB}$$

$$(\varepsilon_p\omega) \sim 4(\varepsilon_p\omega)_{D,1} + (\varepsilon_p\omega)_{AB}$$

In order to prepare the lead absorbers, a large sheet of lead (density of 11.35 g/cm^3) of thickness 2 mm was used as the base material. This sheet was carefully cut into smaller plates, each with dimensions of 6.5 cm by 6.5 cm and thickness of about 2 mm. A total of 13 plates were prepared from the original sheet.

In the experiment, the ^{152}Eu point source was placed 7 cm away from the detector. The gamma-rays ranging from 121 keV to 1400 keV fall on the Clover detector, and the spectrum has been recorded using the NIAS MARS software for ≈ 3 minutes. Initially, the gamma-ray spectrum was recorded without any lead plate between the source and the detector. Subsequently, lead plates were introduced one by one between the source and the detector, and the gamma-ray spectrum was recorded after the addition of each plate. The attenuation of gamma rays in a material follows the exponential attenuation law, given by;

$$I = I_0 e^{-\mu x}$$

where I_0 is the initial intensity of the gamma ray without any absorber, I is the transmitted intensity after passing through the absorber, μ is the linear attenuation coefficient of lead, and x is the thickness of the lead absorber. Taking the natural logarithm of both sides of the above equation, it becomes as;

$$\ln(I) = -\mu x + \ln(I_0)$$

For the sake of convenience, the equation can also be written in terms of mass thickness. Since x can be expressed as ρx, where ρ is the density of lead, the equation becomes;

$$\ln(I) = -\left(\frac{\mu}{\rho}\right)(\rho x) + \ln(I_0)$$

The above expression is similar to the straight-line equation $y = mx + c$. Therefore, if $\ln(I)$ is plotted against the mass thickness ρx, a straight line is obtained. The slope of this straight line is equal to $-\mu/\rho$. As such, μ/ρ represents the mass attenuation coefficient of lead, and ρx represents the mass thickness of the absorber.

In the experiment, gamma-ray intensities were obtained from the recorded energy spectra using the clover HPGe detector. For each gamma-ray energy peak, the intensity I was measured for different thicknesses of lead absorbers placed between the source and the detector. A graph was then plotted between $\ln(I)$ and the mass thickness ρx. According to the above equation, this plot gives a straight line. The slope of the straight line is equal to $-\mu/\rho$. From the slope, the mass attenuation coefficient was determined. The linear attenuation coefficient μ may then be obtained by multiplying the mass attenuation coefficient by the density of lead. As a representative case, the two plots at energies 344.28 MeV and 778.92 MeV gamma rays as a function of absorber thickness in units of mass thickness (g/cm^2) are shown in Fig. 2. The slope of the graph gives the mass attenuation coefficient of lead at each energy. In this way, the mass attenuation coefficients were determined for each identified gamma ray of the ^{152}Eu source.

In addition to attenuation (mass absorption coefficient) measurements, the efficiency of the HPGe clover detector was also determined. The efficiency of the detector for each gamma ray energy was deduced using the expression;

$$\varepsilon = \frac{N}{N_o e^{-\lambda t}\theta}$$

Here, 'λ' denotes the decay constant of the radioactive gamma-ray source, 't' is the time elapsed between the source manufacturing date and the date on which the experiment was performed, and 'θ' represents the branching ratio of the gamma ray. Using the above expression, the direct, add back and total efficiency of the detector have been deduced.

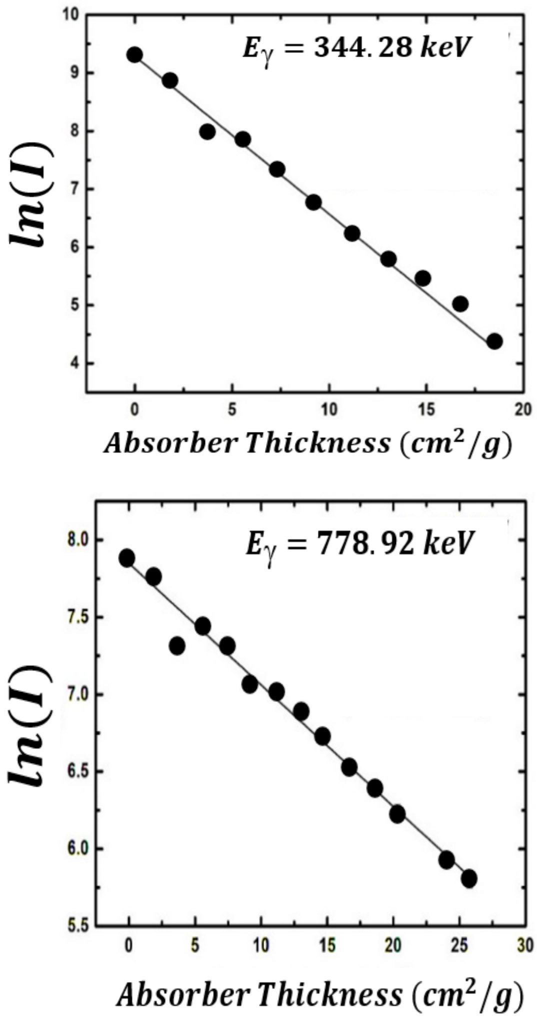


Figure 2: *$ln(I)$ vs. mass thickness (ρx) plot for two gamma-ray energies (344.28 keV and 778.92 keV).*

Results and Discussion

In the present work, the mass attenuation coefficients (μ/ρ) of lead were determined for gamma rays of different energies. The experimentally measured values are plotted as a function of incident gamma-ray energy, as shown in Fig. 3(a). This nicely demonstrates the energy dependence of gamma-ray attenuation in lead. The results show a decreasing trend of μ/ρ with increasing photon energy, as expected.

The efficiency characteristics of the detector system were also examined in detail. To illustrate this, a composite graph has been plotted, as shown in Fig. 3(b). This graph shows the variation of geometry-dependent efficiency for each of the four operating crystals, the combined add-back efficiency, and the overall total efficiency of the detector system as a function of gamma-ray energy.

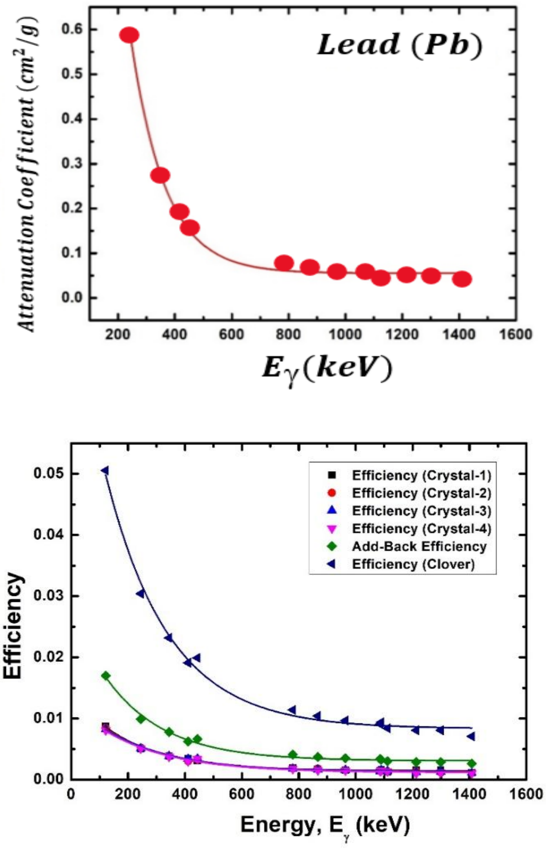


Figure 3: *Variation of mass attenuation coefficient (upper panel) and efficiency of detector with incident gamma-ray energy (lower panel).*

This study of the gamma-ray attenuation coefficient is consistent with the theoretical understanding of photon interactions with matter. At lower photon energies, the photoelectric effect dominates, leading to higher attenuation due to its strong dependence on atomic number. As the photon energy increases, the probability of photoelectric absorption decreases, and Compton scattering becomes the dominant interaction mechanism. This transition results in a gradual decrease in the attenuation coefficient, explaining the observed inverse relationship between photon energy and the mass attenuation coefficient.

The efficiency study further indicates that detector efficiency decreases with increasing gamma-ray energy. This decreasing trend occurs because, at low energies, photons readily interact through photoelectric absorption and Compton scattering, resulting in higher detected counts. As the photon energy increases, the interaction probabilities decrease, and a larger number of photons pass through the detector without interaction.

The study also demonstrates the significant role of clover detectors in improving gamma-ray detection efficiency. The use of four crystals leads to an approximately fourfold increase in efficiency. In addition, the add-back technique plays a crucial role in further enhancing the efficiency, resulting in an overall efficiency that is approximately six times higher than that of a single crystal.

Gamma-ray attenuation strongly depends on photon energy. It also depends on the interaction mechanisms involved. Detector design plays an important role in detection efficiency. Data-processing techniques are also crucial. The use of clover detectors improves efficiency. The add-back method further enhances detection efficiency, especially at higher gamma-ray energies.

References

[1] Hubbell, J. H. (1982). "Photon mass attenuation and energy-absorption coefficients" *International Journal of Applied Radiation and Isotopes*, **33**(11), 1269-1290.

[2] Knoll, G. F. (2010). *Radiation Detection and Measurement* (4th ed.). John Wiley & Sons.

[3] Krane, K. S. (1987). *Introductory Nuclear Physics*. John Wiley & Sons.

Mr. Lalit Kumar *completed his M.Sc. in Physics in 2025 from the Department of Physics, Aligarh Muslim University, with Nuclear Physics as a special paper.*

Mr. Aquib Siddique *is a research scholar in the Department of Physics, Aligarh Muslim University, Aligarh.*

Dr. Mohd. Shuaib *is an Assistant Professor in the Department of Physics, Aligarh Muslim University, Aligarh.*

Dr. Mohd. Shariq Asnain *is a faculty member at Jamia Hamdard University, New Delhi.*

NUCLEAR HORIZONS: A Bulletin of Indian Nuclear Physics Association

POSITRON EMISSION TOMOGRAPHY (PET Scan)

R. Prasad and B. P. Singh
Department of Physics, Aligarh Muslim University, Aligarh

Abstract: *The Positron Emission Tomography (PET) is a nuclear technique used in medicine that enables non-invasive imaging of physiological and metabolic processes in the human body. Unlike conventional imaging methods that primarily show anatomical structure, PET provides functional information by tracking the distribution of short-lived radioactive tracers. In PET, positron-emitting radionuclides such as fluorine-18, commonly administered as fluorodeoxyglucose (^{18}F-FDG), participate in normal metabolic pathways. The emitted positrons annihilate with electrons, producing pairs of 511 keV gamma photons that are detected in coincidence. Regions with abnormally high tracer uptake, such as cancerous tissues or malfunctioning brain areas, appear as metabolic hotspots. By combining nuclear physics principles with advanced detector systems and image reconstruction algorithms, PET has become an indispensable tool for diagnosis, staging, therapy monitoring, and biomedical research.*

Introduction

Positron Emission Tomography, or PET, is an incredible imaging technique that allows us to "see" how different parts of the body are functioning, not just how they look. Instead of focusing solely on anatomy, PET captures the activity happening inside our tissues, revealing the subtle metabolic and biological processes that keep us alive and well. Positron Emission Tomography gets its name from both the underlying nuclear process and the imaging method. The term "tomography" comes from two Greek words: "tomo", meaning slice or section, and "graphy", meaning to write or create an image. In PET, radioactive tracers inside the body emit positrons, which annihilate with electrons to produce pairs of 511 keV photons. These photons are detected in coincidence, and mathematical reconstruction techniques are used to "graph" or generate images in "tomo" form, slice-by-slice cross-sections of metabolic activity. Thus, the name reflects its core principle: imaging the body by creating sectional pictures based on positron emission.

Think about what happens when we eat. The food we consume, whether it's carbs, proteins, or fats, is broken down into glucose, amino acids, and fatty acids. These nutrients enter our bloodstream and are carried to every corner of the body. Once they reach our cells, they are transformed into energy, the fuel that powers everything we do. Energy is essentially the ability to do work, and it is this transformation that has allowed humans to build civilizations, invent technologies, and accomplish amazing feats. Both energy and work are measured in joules (J), though other units exist as well.

The body's process of converting food into usable energy is called metabolism. Metabolism is constantly at work, providing the energy needed for essential functions like breathing, pumping blood, repairing tissues, regulating hormones, maintaining body temperature, and digesting food. Even when we're resting, our bodies require a baseline amount of energy to survive, this is known as the Basal Metabolic Rate (BMR). It's fascinating to note that BMR varies from person to person. On average, a man requires about 7,100 kilojoules per day, while a woman needs around 5,900 kilojoules.

PET scans take advantage of these metabolic processes. By tracking how the body uses nutrients like glucose, PET can highlight areas of unusually high or low activity, helping doctors detect diseases, monitor organ function, or evaluate the effectiveness of treatments. In essence, PET lets us peer into the living engine of the body, revealing its dynamic energy patterns in real time.

Abnormal parts of the body, such as those affected by cancer, behave very differently from healthy tissue. One striking difference is in how they handle energy. Normal cells have a balanced way of converting carbohydrates, fats, and proteins into energy. Cancer cells, on the other hand, are "greedier", they consume far more glucose just to produce the same amount of energy. This means that areas with cancer have a much higher glucose uptake compared to healthy tissue. Similarly, in the brain, regions that aren't functioning properly can show increased metabolic activity. In short, when a part of the body is abnormal, it tends to hoard sugar, creating a hotspot of metabolic activity.

This is exactly what PET scans take advantage of. During a PET scan, a patient is given a special type of sugar, usually glucose, tagged with tiny, short-lived radioactive atoms. This "tracer-labelled sugar" acts like a beacon inside the body. It is usually injected into a vein through an IV line, and within 30 to 90 minutes, it travels through the bloodstream, accumulating in the parts of the body with unusually high metabolic activity.

Once the tracer collects in these hotspots, the radioactive atoms begin to decay, emitting radiation. The PET scanner is equipped with sensitive detectors that capture this radiation and send the information to a computer. The computer then processes the data and produces detailed images, highlighting the abnormal areas where the tracer has piled up. The brighter or more intense the image, the higher the metabolic activity, and the more likely it is that the tissue is abnormal.

Radioactivity and Tracers: A Window into the Invisible

Some atoms in nature are a bit "unstable." Over time, they spontaneously break apart, releasing energy in the process. This natural tendency to disintegrate is what we call radioactivity, and it's a phenomenon that happens right in the heart of the atom, the nucleus. A key concept in radioactivity is the half-life, which is the time it takes for half of a sample of radioactive atoms to decay. Depending on the element, half-lives can range from billions of years to just a few millionths of a second.

Scientists have also learned how to create radioactive atoms in the lab. By bombarding stable atoms with high-energy particles like protons, neutrons, or alpha particles, accelerated in giant machines called particle accelerators, we can produce new radioactive isotopes. What makes these atoms particularly useful is that each type of radioactive atom decays in a unique way, emitting specific forms of radiation such as alpha particles, beta particles, or gamma rays. This "signature decay" allows scientists to identify and track these atoms with great precision.

In medical and biological research, we often use radioactive tracers—tiny amounts of short-lived radioactive atoms mixed with substances that the body naturally processes, like glucose. Once injected into the body, these tracers follow the natural pathways of metabolism. By detecting the radiation emitted as these atoms decay, researchers and doctors can visualize how substances move through tissues, organs, and even entire organisms. The beauty of using short-lived tracers is that they decay quickly, so almost all the radioactive atoms disappear from the body in a matter of hours, leaving little to no lasting radiation.

In PET scanning, the most commonly used tracer is an isotope of fluorine called ^{18}F, combined with glucose to form fluorodeoxyglucose (FDG). ^{18}F has a half-life of about 110 minutes, meaning it essentially leaves the body within about 10 hours. When ^{18}F decays, it emits a tiny particle called a positron (β^+), the antimatter twin of the electron. While an electron carries a negative charge, a positron carries an equal positive charge. Positrons are just one example of the fascinating world of particle-antiparticle pairs in physics, but in PET scans, they play a starring role: when a positron meets an electron in the body, they annihilate, producing gamma rays that the PET scanner can detect, allowing doctors to map the body's metabolic activity with incredible precision.

Annihilation Gamma Rays: How PET Sees Metabolic Hotspots

Abnormal parts of the body, like cancerous tissue or regions of the brain that aren't functioning properly, have much higher metabolic activity than healthy tissue. This means that when a patient is injected with ^{18}F-labeled glucose (FDG) through an IV, a larger portion of this radioactive sugar naturally accumulates in the abnormal areas. Once the tracer builds up, the ^{18}F atoms begin to decay, and each decay releases a tiny particle called a positron (β^{+}). This positron doesn't travel far, typically less than 1 millimetre, before it collides with an electron (β^{-}), which is always present in the surrounding atoms. When a positron meets an electron, something remarkable happens: they annihilate each other.

During annihilation, the positron and electron disappear, converting their mass into energy, and in the process, two gamma rays are emitted in opposite directions. Each of these gamma rays carries an energy of 511 keV. If the positron and electron meet while completely at rest, the gamma rays fly apart at a perfect 180° angle. In reality, though, they're often still moving when they collide, so the angle can be slightly less, around 178°.

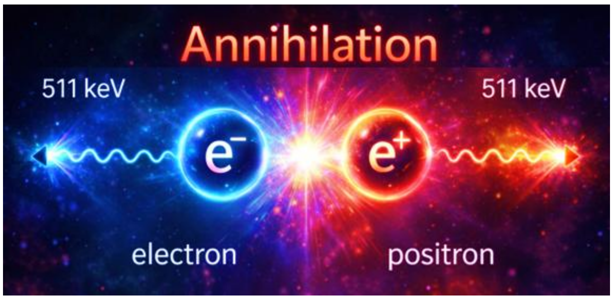


Figure 1: *Pictorial representation of annihilation of electron-positron leading to two gamma photons of 511 keV each*

These gamma rays are what the PET scanner detects. By tracking their origin and paths, the scanner's computer can pinpoint exactly where the tracer has accumulated, effectively highlighting the "hotspots" of abnormal metabolic activity in the body. In other words, PET doesn't just take a static picture—it maps the very energy processes of living tissue, revealing areas where cells are unusually active. (In diagrams, the size of atoms, electrons, and the distances involved are often greatly exaggerated to make the process easier to visualize.)

How PET Detectors Capture the Invisible

Inside a PET scanner, pairs of gamma ray detectors sit directly opposite each other, ready to catch the tiny flashes of energy produced when positrons and electrons annihilate. These detectors are finely tuned to pick up 511 keV gamma rays, the exact energy released during this annihilation, and to record them simultaneously within an incredibly short window of a billionth of a second (10^{-9} seconds). By looking only for gamma rays that meet these strict criteria, the system effectively ignores stray background radiation, ensuring that what's being measured comes specifically from the positron-electron annihilation events.

When the detectors capture these gamma rays, they send the signals to the PET scanner's computer. Using sophisticated software, the computer traces the paths of the gamma rays backward, pinpointing the exact location in the body where the annihilation occurred. The more tracer that has accumulated in a particular area, like an abnormal or cancerous tissue, the more annihilation events happen there. The computer collects all this information and "illuminates" these hotspots, creating a detailed map of metabolic activity in the body.

Interestingly, in the final PET images, the areas of high tracer concentration often appear darker, as the images are essentially negative representations of the actual activity. In essence, PET scanning allows

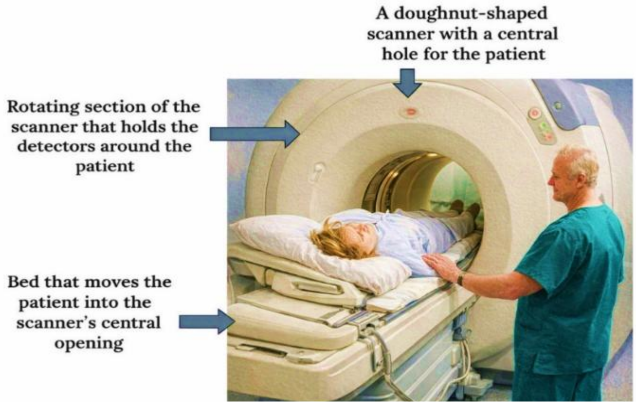


Figure 2: *PET scanner in operation with the patient inside the scanning tunnel*

doctors to map the three-dimensional distribution of the tracer inside the body, revealing where cells are unusually active and giving insights that go far beyond conventional imaging techniques.

PET Scanner unit

A PET scanner may look like a piece of futuristic machinery, and in many ways, it is. The system is built around four major components that work together with precision. The control console serves as the command centre, where the technologist operates the scanner, adjusts settings, and monitors the entire procedure. The minicomputer, equipped with advanced reconstruction software, processes the millions of tiny signals detected during the scan and converts them into high-quality, clinically useful images.

At the core is the large, doughnut-shaped gantry, a circular ring into which the patient is moved on a motorized bed. The gantry can rotate smoothly around the patient, collecting data from multiple angles. Inside this ring are several detector modules arranged in a full 360-degree array. These detectors, made of scintillation crystals coupled with photodetectors, pick up pairs of gamma rays produced when a positron emitted from the radiotracer annihilates with an electron in the body. Modern PET scanners often include multiple detector rings, allowing faster acquisition and higher sensitivity. Working together, these components allow the PET system to construct detailed 3D maps of metabolic and biochemical activity inside the body, transforming invisible molecular processes into clear images that guide diagnosis, staging, and treatment planning.

Each PET detector setup is made up of a pair of gamma ray detectors positioned directly opposite to each other, designed to catch the twin gamma rays produced when a positron from the tracer meets an electron in the body. When these detectors record gamma rays simultaneously, the system's analysing software springs into action, tracing the paths of the gamma rays to pinpoint exactly where in the body the annihilation occurred and highlighting that spot. During a scan, as the scanner rotates around the patient, multiple "snapshots" are taken of the area directly beneath the scanner, capturing regions where the FDG tracer has concentrated. The software then combines these snapshots into a single image called a slice. By moving the patient's bed and scanning successive sections of the body, the system can generate multiple slices, which are then stitched together to produce a complete, full-body image.

Full-body PET scans are incredibly valuable for identifying regions of unusually high metabolic activity, which could indicate early-stage cancer, pre-cancerous developments, or other abnormal metabolic changes. To increase accuracy, PET findings are often combined with CT scans, providing both functional and anatomical insights. One of the greatest advantages of PET scanning is its safety and non-invasive nature. The radioactive tracer is used in very small amounts and has a short half-life, meaning it decays and leaves the body within a few hours. Positrons themselves vanish within nanoseconds as they meet electrons, so there's no lingering effect on the patient or those around them. This makes PET scanning a safe and effective way to visualize the body's inner workings in real time.

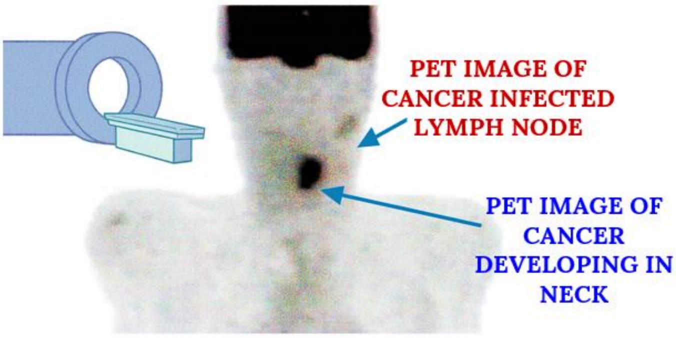


Figure 3: *FDG-PET scan of the neck demonstrating increased uptake in a diseased lymph node and another abnormal site.*

Modern PET scanning has improved a lot in recent years. New digital detectors give clearer and faster images. Small tumours and early disease changes can now be seen more easily. Time-of-Flight (TOF) technology improves image quality by more accurately locating the source of positrons. New radiotracers let doctors' study specific body functions, not just cancer. PET can be combined with CT or MRI for detailed pictures of structure and function. AI helps make scans faster, safer, and more accurate. These advances make diagnosis and treatment planning better for patients.

In summary, PET scanning opens a window into the body's hidden world, allowing us to see not just what organs look like, but how they function at a cellular level. By following the journey of tiny, safe radioactive tracers through the bloodstream, doctors can spot areas where cells are unusually active, often long before structural changes appear. This remarkable ability to visualize metabolism in real time provides a powerful tool for early diagnosis, treatment planning, and monitoring the effectiveness of therapies. PET scans showcase how the principles of nuclear physics are not just abstract concepts in a lab. They have practical, life-saving applications in modern medicine, turning invisible processes into actionable insights that guide patient care.

Prof. R. Prasad *has been a distinguished faculty member in the Department of Physics, Aligarh Muslim University (AMU), Aligarh. His research interests include nuclear reaction dynamics, heavy-ion collisions, pre-equilibrium and compound nuclear processes, and nuclear structure effects in nuclear reactions. He is a prolific writer, and has written a large number of books, with a large number of research publications in leading national and international journals. Prof. Prasad has made significant contributions to nuclear physics research and is highly regarded for his role as an excellent teacher, in mentoring and guiding young researchers.*

Prof. B. P. Singh *is Senior Professor and a faculty member in the Department of Physics, Aligarh Muslim University (AMU), Aligarh.*

New-generation gamma detector arrays for nuclear physics research and societal applications

Samit K. Mandal
Department of Physics & Astrophysics, University of Delhi, Delhi 110007, India
e-mail: smandal@physics.du.ac.in

Abstract: *Gamma-ray spectroscopy has been a central experimental tool in nuclear physics since the earliest studies of radioactive decay. Progress in detector technology has repeatedly transformed the scope and precision of nuclear structure research, enabling deeper insight into collective motion, shell evolution and exotic modes of excitation. From early gas-filled detectors and scintillators to high-purity germanium detectors and, more recently, to highly segmented detectors employing digital signal processing, pulse-shape analysis and gamma-ray tracking, each technological advance has addressed specific experimental challenges. In the present era of radioactive ion beam facilities, experiments are increasingly limited by low production rates, high background and large Doppler effects, demanding detector arrays with very high efficiency, excellent energy and position resolution and strong background suppression. This invited general review presents a unified and compact account of the evolution of gamma-ray detector technology, combining recent developments in digital pulse processing, gamma-ray tracking and imaging-capable detector concepts. Particular emphasis is placed on the DGAS (DESPEC Germanium Array Spectrometer) initiative and related efforts aimed at next-generation nuclear spectroscopic research. Contributions from the University of Delhi group to pulse-shape analysis, gamma tracking and collaborative detector development are highlighted, together with broader societal applications of advanced gamma-detection technologies.*

Keywords: Gamma-ray spectroscopy; HPGe detectors; digital signal processing; pulse-shape analysis; gamma tracking; gamma imaging; DGAS.

Gamma rays emitted in nuclear transitions provide direct information on the quantized energy levels, angular momentum and intrinsic structure of atomic nuclei. As a consequence, gamma-ray spectroscopy has been inseparable from the development of nuclear physics itself. In the earliest experiments, gamma radiation was detected using gas-filled devices such as ionization chambers and Geiger–Müller counters. While these detectors were robust and adequate for establishing the existence and basic properties of ionizing radiation, they offered no meaningful energy resolution and therefore limited spectroscopic capability. The introduction of scintillation detectors in the mid-twentieth century marked the first major step towards practical gamma-ray spectroscopy. Inorganic scintillators, notably NaI(Tl), provided high detection efficiency and moderate energy resolution, enabling the construction of early nuclear level schemes and systematic studies of radioactive decay. However, the intrinsic statistical limitations of scintillation light production restricted their resolving power, particularly for complex nuclei with closely spaced transitions.

A decisive breakthrough occurred with the advent of semiconductor detectors. Lithium-drifted germanium detectors introduced in the 1960s offered an order-of-magnitude improvement in energy resolution over scintillators, transforming gamma-ray spectroscopy into a precision tool. This development was followed by the emergence of high-purity germanium (HPGe) detectors, which combined excellent energy resolution with operational stability at liquid-nitrogen temperatures. HPGe detectors rapidly became the standard for nuclear structure studies. Their superior resolution enabled the identification of detailed level schemes, rotational and vibrational band structures, shape coexistence phenomena and subtle single-particle effects. Coincidence techniques using multiple HPGe detectors further enhanced experimental capability by allowing the reconstruction of complex decay cascades and the extraction of angular correlation and polarization information. Composite detector systems such as clover detectors, employing add-back techniques and Compton suppression, significantly improved efficiency and peak-to-background ratios.

As nuclear spectroscopy advanced, the need for higher efficiency and angular coverage led to the development of large gamma-ray detector arrays. Arrays such as GAMMASPHERE and EUROBALL, consisting of dozens to hundreds of Compton-suppressed HPGe detectors arranged in near-4π geometries, represented the culmination of this approach. These instruments enabled high-fold gamma–gamma coincidence measurements and provided unprecedented resolving power for studies of high-spin phenomena and collective excitations. In India, the establishment of the Indian National Gamma Array (INGA) similarly strengthened national capabilities in discrete gamma spectroscopy. Despite their success, conventional arrays based on escape-suppressed detectors face intrinsic limitations. A significant fraction of the solid angle is occupied by passive shielding, limiting photopeak efficiency to around ten per cent, while Compton scattering between detectors and summing effects become increasingly problematic at high gamma multiplicities.

The emergence of radioactive ion beam facilities has further exposed these limitations. Experiments on exotic nuclei are often characterized by low beam intensities, high background from atomic processes and large Doppler shifts and broadening due to high recoil velocities. Under such conditions, maximizing efficiency, granularity and Doppler correction capability becomes essential. These challenges motivated the development of a fundamentally new concept in gamma-ray detection, namely gamma-ray tracking, which exploits recent advances in detector segmentation, digital electronics and signal processing.

In highly segmented HPGe detectors, a gamma ray typically undergoes multiple interactions through Compton scattering and photoelectric absorption. Each interaction produces charge carriers whose drift induces characteristic current signals not only on the segment where the interaction occurs but also, through capacitive coupling, on neighbouring segments. The detailed time structure of these real and mirror signals contains information about the three-dimensional position of each interaction within the detector volume. Pulse-shape analysis techniques aim to extract this information by comparing measured signals with reference pulse shapes obtained from detector scanning or from detailed simulations based on electric-field and weighting-field calculations. Considerable effort has been devoted worldwide to the development of robust pulse-shape analysis algorithms, including methods based on signal deconvolution, pattern recognition, artificial neural networks and wavelet transforms.

The implementation of fully digital signal processing, in which preamplifier outputs are sampled by fast analogue-to-digital converters, has been crucial for the practical realization of pulse-shape analysis. Digital electronics provide excellent stability, precise timing, flexibility in triggering and the ability to implement sophisticated algorithms either offline or in real time. Once the positions and energies of individual interaction points are determined, gamma-ray tracking algorithms reconstruct the full interaction sequence of each gamma ray. By applying Compton kinematics and statistical criteria to all possible permutations of interaction points, the most probable sequence is identified, allowing reconstruction of the original gamma-ray energy and emission direction. Two main classes of algorithms, clusterization and backtracking, are widely employed and continue to be refined.

The advantages of gamma-ray tracking are substantial. By replacing passive Compton suppressors with active germanium, tracking arrays maximize the solid angle covered by high-resolution detectors, leading to photopeak efficiencies approaching fifty per cent. Knowledge of the first interaction point enables precise Doppler correction even for fast-moving nuclei, while the ability to reject partially absorbed events significantly improves peak-to-background ratios. In addition, tracking provides access to gamma-ray polarization information and, when combined with suitable detector geometry, intrinsic imaging capability. These features make gamma-ray tracking particularly well suited for experiments at radioactive ion beam facilities, where efficiency and background suppression are critical.

The most advanced realizations of this concept are the Advanced GAmma Tracking Array (AGATA) in Europe and the Gamma-Ray Energy Tracking Array (GRETA) in the United States. Both projects employ highly segmented HPGe detectors arranged in modular geometries and rely on sophisticated pulse-shape analysis and tracking software. Extensive experimental campaigns and simulations have demonstrated their superior performance compared to previous-generation arrays, establishing gamma-ray tracking as the dominant paradigm for future high-resolution gamma spectroscopy.

Building on this international experience, efforts are underway in India to develop expertise and infrastructure for next-generation gamma-tracking arrays. A central element of these efforts is the DGAS (DESPEC Germanium Array Spectrometer) initiative, a collaborative programme aimed at developing an image-capable gamma-tracking array tailored to the requirements of future nuclear spectroscopy experiments. DGAS emphasizes modularity, high granularity and advanced digital processing, with particular focus on imaging capability for experiments involving implanted radioactive nuclei and isomeric decays. Such capability is especially relevant for decay spectroscopy at radioactive beam facilities, where precise localization of gamma-ray emission is essential for background suppression and event correlation.

The University of Delhi group has played a significant role in the development of pulse-shape analysis and gamma-tracking activities in India. The group has established dedicated facilities for the characterization of segmented planar HPGe detectors, including collimated source scanning systems to generate reference pulse-shape databases. Substantial effort has been devoted to the development and optimization of pulse-shape analysis algorithms, achieving millimeter-scale position resolution, which is a prerequisite for effective gamma tracking. In parallel, the group has contributed to simulation studies and algorithm development for tracking, exploring both clusterization and backtracking approaches.

These activities are closely linked to national and international collaborations. The University of Delhi group has been actively involved in the HISPEC and DESPEC programmes at the FAIR–NUSTAR facility in Germany, contributing to detector development, electronics and software. Collaboration with Indian laboratories such as TIFR, VECC, IUAC, SINP and several Universities has further strengthened national capabilities, while sustained interaction with international groups has ensured alignment with global developments. Through these efforts, the group has established a visible presence in the international gamma-detector community and has helped position India as an active participant in next-generation detector development.

While the primary motivation for gamma-tracking arrays lies in fundamental nuclear physics, the technological advances underpinning these systems have significant societal impact. Gamma imaging and tracking concepts developed for nuclear spectroscopy are directly relevant to medical imaging techniques such as positron emission tomography and single-photon emission computed tomography, where precise localization of gamma-ray sources is crucial. High-resolution portable gamma spectrometers based on germanium detectors are widely used in radiation safety, environmental monitoring and homeland security for the identification of radioactive sources. In industry, advanced gamma detectors enable non-destructive testing and materials analysis. Thus, investments in fundamental detector research yield broad technological benefits beyond the laboratory.

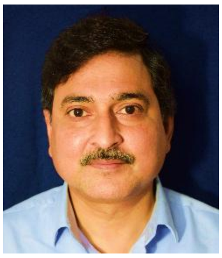
In summary, the evolution of gamma-ray detector technology from simple gas-filled counters to sophisticated image-capable tracking arrays has been a driving force in nuclear physics. New-generation gamma detector arrays based on segmented HPGe detectors, digital electronics, pulse-shape analysis and gamma-ray tracking are essential for addressing the challenges posed by modern nuclear structure and nuclear astrophysics research, particularly at radioactive ion beam facilities. The DGAS initiative represents an important step towards developing indigenous capability in image-capable gamma tracking, building on expertise developed through national and international collaboration. Continued progress in this field promises not only deeper insight into the structure of atomic nuclei but also sustained societal benefits through applications of advanced gamma-detection technologies.

Acknowledgements

The author acknowledges fruitful collaborations with national partners (TIFR, VECC, IUAC, SINP) and international collaborators (GSI–FAIR) and support from funding agencies including the Department of Science and Technology (DST), Government of India, and the Institution of Eminence (IoE), University of Delhi, which have enabled detector development and related research activities.

References

1. Knoll G. F., *Radiation Detection and Measurement*, 4th edn, Wiley, New York, 2010.
2. Eberth J. and Simpson J., Prog. Part. Nucl. Phys., 2008, 60, 283–337.
3. Akkoyun S. *et al.*, Nucl. Instrum. Methods Phys. Res. A, 2012, 668, 26–58.
4. Michelagnoli C. *et al.*, Nucl. Instrum. Methods Phys. Res. A, 2013, 729, 334–349.
5. Grebosz J. *et al.*, Nucl. Instrum. Methods Phys. Res. A, 2020, 959, 163475.
6. Lee I. Y., Johnson N. J. and Ragnarsson I., Nucl. Phys. A, 2004, 746, 255–259.
7. Sharpey-Schafer J. F., Prog. Part. Nucl. Phys., 1988, 21, 293–400.
8. Beausang C. W. and Frauendorf S., J. Phys. G: Nucl. Part. Phys., 1996, 22, 527–558.
9. Svensson C. E. *et al.*, J. Phys. G: Nucl. Part. Phys., 2005, 31, S1663–S1670.
10. Satpathy L., Nayak R. C. and Patra S. K., J. Phys. G, 2004, 30, 771–785.
11. Sheikh J., Palit R. and Satuła W., Phys. Rev. C, 2005, 72, 041301.
12. Siwal D. *et al.*, Rev. Sci. Instrum., 2013, 84, 075117; Nucl. Instrum. Methods Phys. Res. A, 2014, 741, 108–116.
13. Banerjee A. *et al.*, Nucl. Instrum. Methods Phys. Res. A, 2019, 930, 100–104.
14. AGATA Collaboration, Nucl. Instrum. Methods Phys. Res. A, 2012, 668, 26–58.
15. FAIR–NUSTAR Collaboration, Technical Design Reports for HISPEC, DESPEC and DEGAS detector systems, FAIR Reports, Darmstadt, 2011–2018.



Prof. Samit K. Mandal *is a Senior Professor in the Department of Physics & Astrophysics, University of Delhi, Delhi. His research focuses on nuclear reaction dynamics and nuclear structure studies, with extensive experience in radioactive ion beam facilities, advanced detector systems, and large international collaborations such as RISING and AGATA at GSI, Germany.*

INPA News & Announcements

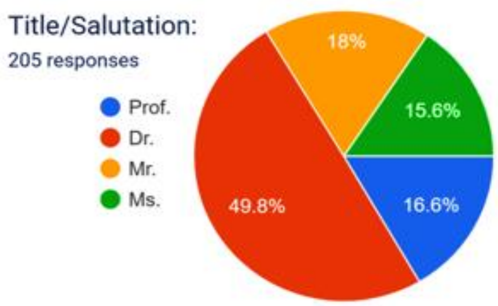
Membership updates, upcoming events, workshops, and collaboration initiatives

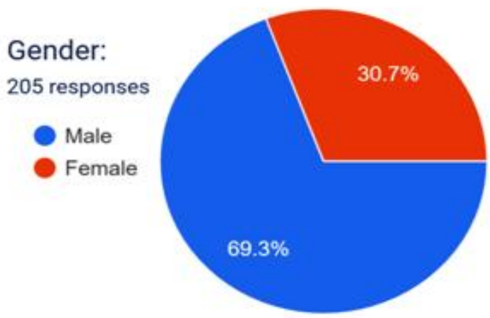
Following the INPA meeting held on 5 November 2025, a Google Membership Form was launched to create a unified and vibrant platform for the Indian nuclear physics community. The response has been extremely encouraging. As of today, 205 members have registered, representing a wide and diverse spectrum of contributors from across the country—and even beyond.

A Snapshot of the Membership Profile

A careful analysis of the received entries highlights a balanced and impressive participation from all segments of the nuclear physics ecosystem. A significant proportion of the 205 members consists of Research Scholars and PhD students from AMU, BHU, DU, IITs, IISERs, Central Universities, TIET Patiala, Panjab University, University of Calicut, Bose Institute, NIET, and several others. This strong representation of young researchers reflects a healthy pipeline of future scientists and demonstrates the enthusiasm of the next generation to engage with the national community.

A substantial number of Professors, Associate Professors, Assistant Professors, and Emeritus Academicians have joined from prominent institutions such as IIT Roorkee, University of Delhi, AMU, Panjab University, Bharathiar University, Visva-Bharati, Allahabad University, and many others. Their presence ensures academic depth, mentorship, and a robust leadership base for INPA. Members from national research centres: BARC, VECC, IUAC, IOP Bhubaneswar, NAPS (NPCIL), are actively part of the association. Their participation strengthens the link between fundamental nuclear physics and national research programmes, bringing rich laboratory experience to the community. Several Assistant Professors and early-career faculty from universities, autonomous colleges, and engineering institutes have registered. Their inclusion supports INPA's mission to promote research-driven teaching and expand nuclear physics activities across varied educational settings.





A notable number of Indian researchers currently working abroad, at GSI Darmstadt (Germany), the Hebrew University of Jerusalem (Israel), University of Zagreb (Croatia), and Brno University of Technology (Czech Republic), have also joined. Their involvement demonstrates their continued connection with the Indian community and adds valuable international exposure to INPA's network. The analysis of the first 205 members clearly reveals:

- Research Scholars form the largest constituency, showing strong youth engagement.
- Senior professors and scientists provide depth, leadership, and continuity.
- Representation spans nearly every major university, IIT, IISER, and national laboratory in India.
- Indian researchers working abroad enrich the association with global perspectives.

With such a dynamic and diverse membership, INPA is poised to move forward with impactful initiatives, advanced schools, thematic workshops, collaborative research meetings, and nationwide outreach activities—to strengthen and expand nuclear physics research and education across India.

NEWS ITEMS

India's Nuclear Physics Community Comes Together: INPA Launched

"A national platform to strengthen nuclear physics research, education, and infrastructure in India"

November 5, 2025
In a significant step towards strengthening nuclear physics research and education in India, the **Indian Nuclear Physics Association (INPA)** was formally launched through an online meeting held on November 5, 2025. The initiative brings together researchers, teachers, students, and professionals from across the country under a common national platform.

India has a strong legacy in nuclear science, with contributions spanning energy, healthcare, industry, and environmental monitoring. However, the nuclear physics community has been facing several challenges, including limited access to accelerator facilities, declining student engagement, and inadequate experimental infrastructure in universities. INPA has been conceived to collectively address these issues and to enhance the visibility and impact of nuclear physics in the national scientific landscape.

The launch meeting witnessed participation from more than 100 scientists and academicians from universities and national laboratories across India. Prof. A. K. Jain was unanimously appointed as the President of INPA, while Prof. B. P. Singh was appointed as the Vice President and Executive Secretary. Senior members of the community emphasized that the formation of INPA was a long-felt need.

Highlighting the significance of the initiative, Prof. A. K. Jain remarked, *"Nuclear science is not only about atoms—it is about nation-building. A national association like INPA can play a transformative role in shaping research and education in this field."*

During the meeting, the vision, mission, and roadmap of INPA were outlined. The association aims to promote both fundamental and applied nuclear research, improve access to accelerator and experimental facilities for university users, support human resource development, and foster collaboration between academia, national laboratories, industry, and policymakers.

Participants also stressed the importance of strengthening hands-on training for students and expanding experimental opportunities at existing facilities such as IUAC, RRCAT, TIFR, and VECC. The INPA is expected to act as a strong collective voice in advocating for improved infrastructure and policy support.

With an inclusive membership structure and a focus on national priorities, INPA seeks to build a vibrant and connected nuclear physics community in India. The coming months will see the formal registration of the association, expansion of membership, and initiation of activities aimed at strengthening research, education, and outreach.

The formation of INPA marks a new chapter for nuclear physics in the Indian university system—one rooted in collaboration, shared vision, and national responsibility.

The ICNSNR-2025 and the Role of INPA in Advancing Nuclear Physics

5-7 May, 2025

On behalf of the organizers, I sincerely thank all the participants, organising committee members and advisory committee members for their valuable support in the International Conference on Nuclear Structure and Nuclear Reactions (ICNSNR-2025) held at the Department of Physics, Aligarh Muslim University, during May 5–7, 2025. More than 100 nuclear physicists participated in the conference from various leading institutions.

The deliberations led to key recommendations aligned with the Mega Science Vision-2035 for Nuclear Physics, with emphasis on strengthening research, infrastructure, and human resource development. Special focus was placed on updating nuclear physics–related course curricula and on regularly organizing conferences and workshops on nuclear physics topics to sustain academic interaction. A major recommendation highlighted the active role of the Indian Nuclear Physics Association (INPA) in popularising nuclear science and nuclear techniques for societal benefit. We look forward to your continued guidance in effectively implementing these recommendations.

DAE SYMPOSIUM ON NUCLEAR PHYSICS HELD AT NIT JALANDHAR

8–12 December 2025

The 69th DAE Symposium on Nuclear Physics was held at Dr. B. R. Ambedkar National Institute of Technology (NIT) Jalandhar from 8–12 December 2025. The symposium was sponsored by the Department of Atomic Energy (DAE), Government of India, and brought together scientists, faculty members, and students from universities and national laboratories across the country. The scientific programme included plenary lectures, oral and poster presentations, and thesis presentations, covering a wide range of topics in nuclear structure, nuclear reactions, and instrumentation. The event provided an important platform for scientific exchange, fostered collaborations, and encouraged young researchers, thereby strengthening India's nuclear physics research ecosystem.

INPA Office at BHU

The members of the Indian Nuclear Physics Association (INPA) express their sincere gratitude to Prof. Ajit Kumar Chaturvedi, Hon'ble Vice-Chancellor of Banaras Hindu University, Varanasi, for kindly allowing to host the INPA office at the Department of Physics, Institute of Science, BHU. Appreciation is also extended to Prof. Sanjay Kumar, Director, Institute of Science, Prof. Rajesh Kumar Shrivastava, Dean, Faculty of Science, and Prof. R. K. Singh, Head of the Physics Department, and all the faculty members, including the nuclear physics community at BHU, for their support and cooperation. Special thanks are due to Prof. Ajay Kumar Tyagi for his efforts and coordination in this regard.

Needless to mention, the Department of Physics at BHU has a long and distinguished tradition in teaching and research, particularly in nuclear physics. Over the years, it has produced many eminent scientists who have made significant contributions to the field and brought recognition to the nation.

The approval to establish the INPA office at BHU marks an important step in strengthening academic and research activities specially that of nuclear physics. It is expected to promote collaboration and create greater opportunities for interaction among students, researchers, and faculty members. Further, it will play an important role in creating awareness about nuclear techniques among the general public. The Department of Physics at BHU, in particular, and the University as a whole are expected to play an important role in advancing the activities of INPA in the coming years. This development will strengthen the nuclear physics community and contribute to continued growth of nuclear physics within the country.

Forthcoming Conferences & Calls for Participation

International Conference at Zakopane, Poland during 30th August to 06th September, 2026

The Zakopane Conference on Nuclear Physics, for historical reasons called School, has been organized since 1963 by the Henryk Niewodniczanski Institute of Nuclear Physics of the Polish Academy of Sciences (IFJ PAN) and the Marian Smoluchowski Institute of Physics of the Jagiellonian University. Over the years the school became famous worldwide conference. Nowadays, the Zakopane Conference on Nuclear Physics has a character of a biennial international congress and is one of the major events in Poland, related to the low energy nuclear physics. This year a special attention will be given to: i) hot topics of the nuclear structure theory including: ab initio calculations, nonequilibrium phenomena and applications of quantum computing, ii) different aspects of nuclear fission, and iii) collective excitations in nuclei. Furthermore, investigations of the properties of exotic nuclei and the nuclear physics context of astrophysical processes will be discussed. The aim of the Conference is also to establish contacts between physicists representing various areas of nuclear physics and to create opportunities for intense interaction between graduate students, young researchers and senior scientists.

Abstract Submission Open for SPaGCoNS 2026

Abstract submissions are now open for the Symposium on Particle–Gamma Coincidence Techniques in Nuclear Science (SPaGCoNS 2026), a three-day scientific symposium that will be hosted by Indian Institute of Technology (Indian School of Mines) Dhanbad from 13–15 July 2026 in Dhanbad, Jharkhand, India. The symposium will bring together researchers, scientists, faculty members, postdoctoral fellows, and students working in the field of nuclear science to discuss recent advances in particle–gamma coincidence techniques, which play a crucial role in modern experimental nuclear physics.

The symposium will focus on several important application areas including nuclear structure, nuclear reactions and reaction mechanisms, nuclear astrophysics, studies involving radioactive ion beams and exotic nuclei, as well as developments in instrumentation and other emerging applications. These topics reflect the growing importance of coincidence measurement techniques in probing nuclear properties and reaction dynamics with greater precision.

The abstract submission portal has now been activated, and the updated abstract template is available on the symposium website. The organizers warmly invite submissions from students, postdoctoral researchers, faculty members, and scientists working in relevant areas. More details about the symposium, submission guidelines, and updates can be accessed at the official website: https://sites.google.com/view/spagcons2026. The organizers look forward to receiving contributions from the scientific community and to welcoming participants to Dhanbad in July 2026.

India's SHANTI Act (2025): A New Chapter for Nuclear Energy and Education

The **SHANTI Act**, officially known as the *Sustainable Harnessing and Advancement of Nuclear Energy for Transforming India Act*, marks a historic reform in India's nuclear energy sector. The Act was passed by Parliament in 2025 and received Presidential assent in late December 2025, making it a legally enforced law. It replaces and modernizes older nuclear regulations to support India's growing energy needs while maintaining strict safety and security standards.

One of the most important features of the SHANTI Act is that it allows private sector participation in nuclear power generation for the first time, under strong government oversight. While strategic and sensitive areas remain under full state control, this change is expected to speed up nuclear power projects, attract investment, and help India meet clean energy and climate goals. Nuclear power is crucial for providing reliable electricity without carbon emissions, especially as energy demand continues to rise. An equally important pillar supporting the success of the SHANTI Act is nuclear education in universities and research institutions. Expanding nuclear power requires a large number of well-trained scientists, engineers, medical physicists, and radiation safety professionals. Universities play a key role in preparing this skilled workforce through courses in nuclear physics, reactor engineering, radiation technology, and nuclear safety. Strong academic programmes also promote research, innovation, and indigenous technology development, which are essential for a self-reliant nuclear sector.

By linking policy reform with education and skill development, the SHANTI Act strengthens the foundation for a safe, peaceful, and sustainable nuclear future. Well-trained graduates from Indian universities will not only support nuclear power plants but also contribute to medical applications, industry, research, and regulation. Together, progressive legislation and strong nuclear education will ensure that India's nuclear expansion remains safe, efficient, and beneficial to society.

Symbol of Science in Service of Society

The emblem of the Indian Nuclear Physics Association represents the spirit of nuclear science community in India, and its contribution to society. The atomic symbol reflects the study of matter and energy, while the surrounding band signifies unity among nuclear scientists. The supporting hand conveys responsible research and peaceful use of nuclear science for the benefit of humanity.

Editorial acknowledgements and concluding remarks

As we conclude this inaugural feature on India's Nuclear Physics Roadmap to 2035, it is important to acknowledge the collective vision, effort, and commitment that have shaped both the national policy framework and the evolving role of the Indian Nuclear Physics Association (INPA). The **Mega Science Vision–2035: Nuclear Physics** document, released by the Office of the Principal Scientific Advisor, stands as a landmark effort that brings clarity, direction, and purpose to the future of nuclear physics in India. This note draws deeply from that vision, and we gratefully recognize the scientific committees, institutions, and experts whose insights have informed the roadmap.

We extend sincere appreciation to the nuclear physics community across universities, research institutes, IITs, IISERs, and DAE laboratories for their continued pursuit of excellence despite numerous challenges. Their work, spanning accelerators, detectors, theoretical modelling, astrophysics, and applied sciences, provides the foundation upon which India's future capabilities will be built. The launch of INPA is itself a testament to the community's determination to unite, strengthen, and advocate collectively for the discipline.

We would like to place on record our heartfelt thanks to all the Professors, Scientists, Researchers, and contributors who have enriched this bulletin with their articles, ideas, and insights. Their thoughtful participation and willingness to share expertise have helped shape Nuclear Horizons into a meaningful platform in its very first issue. We also express our sincere gratitude to the INPA Advisory Committee members and the Executive Committee members for their unwavering support, guidance, and encouragement throughout the preparation of this bulletin. On behalf of the nuclear physics community, I also thank the President of INPA, Prof. A. K. Jain, for his efforts in bringing the community together. Our thanks further extend to all those who have contributed, directly or indirectly, to the development, refinement, and successful release of this inaugural issue of the Bulletin of INPA.

Special thanks are also due to all reviewers and members who provided valuable inputs during the preparation of this volume. Their enthusiasm and engagement reflect a growing momentum toward national coordination and renewed academic vitality.

Looking ahead, the coming decade holds exceptional promise. If the Vision–2035 roadmap is pursued with sustained commitment and strategic investment, India is poised not only to rejuvenate its nuclear physics infrastructure but also to emerge as a global leader in frontier research. The responsibility now lies with all of us, the researchers, educators, students, policymakers, and institutions—to carry this vision forward with collective resolve.

With this, we close the first issue of Nuclear Horizons with optimism, gratitude, and a renewed sense of purpose. The journey toward **India's Nuclear Physics Vision 2050** has truly begun, guided by the collective will of our scientific community and the national aspiration for **Atmanirbhar Bharat**. As we strengthen indigenous capabilities, build world-class facilities, and empower the next generation of researchers, we move steadily toward a self-reliant and globally competitive nuclear science ecosystem. Together, we will shape a brighter, stronger, and more confident scientific future for the nation.

— Editorial Team, Nuclear Horizons

INDIAN NUCLEAR PHYSICS
ASSOCIATION

ADVANCING NUCLEAR SCIENCE
FOR THE BENEFIT OF SOCIETY

"Strengthening India's scientific future through excellence in nuclear physics research, education, and applications."

The Indian Nuclear Physics Association (INPA)
is a national professional body dedicated to promoting research, education, and collaboration in nuclear physics and its applications for societal and strategic development.

✉ **Email:** *inpa.bharat@gmail.com*

An initiative aligned with
Atmanirbhar Bharat & Scientific Self-Reliance